Supplementary Information

Confirmatory evaluation separates uncertainty calibration from discovery performance in materials active learning

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Internal review build. Submission is blocked pending verified metadata.

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S1 Scope and evidence status

This supplement reports the locked V6 confirmation package, its historical provenance, and a separately labelled independent P1 confirmation appendix. V6 pooled inference used 40 paired seed blocks, each averaging the 16 task × split × model environments equally. Environment rows are heterogeneity diagnostics. No new contrast, endpoint, exclusion, environment selection, or statistical analysis was introduced into locked V6 inference.

S2 Evidence lineage and historical provenance

The displays below are historical V2, V3, and V5-RC1 material retained only to document development and method provenance. They are not V6 confirmation evidence, are not pooled with V6, and cannot override the locked V6 results.

Table S1: Historical version and stage roles; no effect estimates are pooled across versions.

Version/stage	Evidence role	Job count	Seed range	Scientific use	Allowed manuscript role	Not allowed to support
V2 historical	exploratory evidence	1,920	2000–2019	method exploration	provenance only	V6 confirmation or cross-version pooled effects
V3 historical	calibration-evaluation evidence	1,920	3000–3039	calibration evaluation	provenance only	V6 confirmation or cross-version pooled effects
V5-RC1 broad development manifest	planned configuration inventory	5,760	1000–1009	strategy inventory	development provenance	confirmatory or effect-estimation claims
V5 reduced selection cohort	executed selection evidence	2,240	1000–1009	policy selection	development and selection provenance	confirmatory or effect-estimation claims
V5-RC1 confirmation	historical locked evidence	3,360	5000–5039	historical locked confirmation	historical evidence and method provenance	V6 confirmation or override of V6
V6 independent confirmation	independent registered confirmation	12,800	7000–7039	frozen trajectory and replay confirmation	V6 confirmatory claims	cross-version pooled effects

Historical/development evidence – not part of V6 confirmatory inference

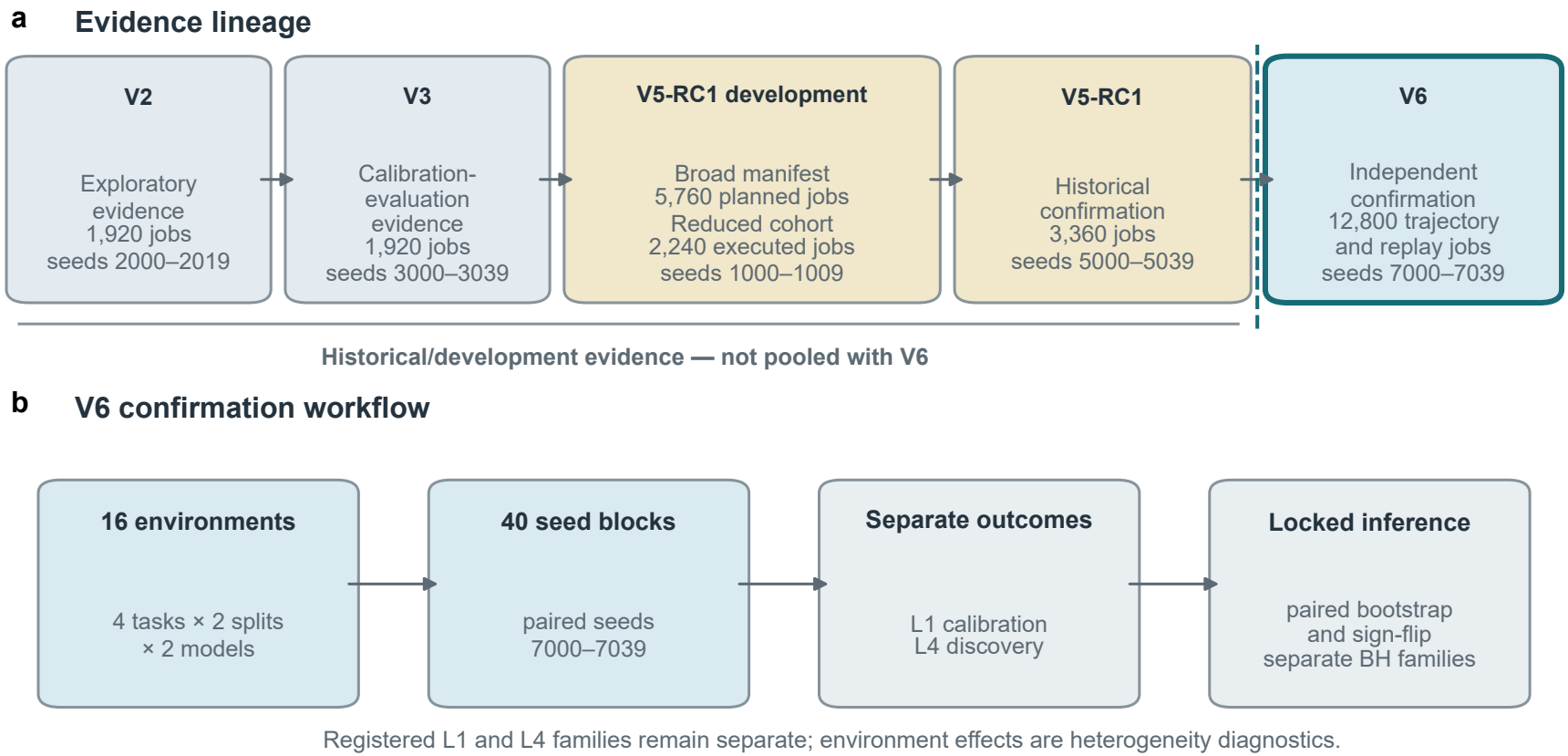


Figure S1: Evidence lineage and V6 confirmation architecture. Historical and development stages motivated but were not pooled with the independent V6 confirmation.

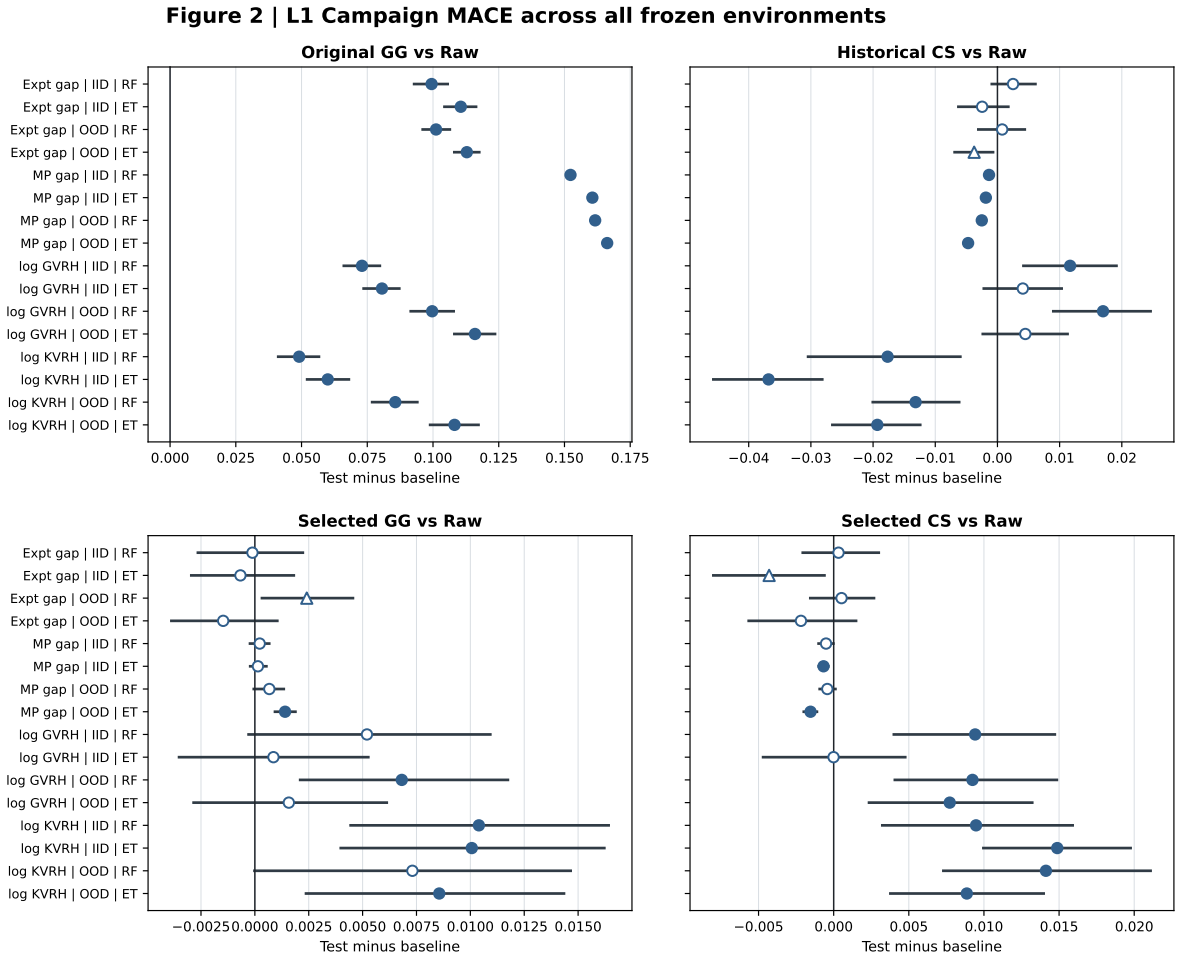


Figure S2: Historical/development evidence – not part of V6 confirmatory inference. V5 L1 environment-level effects.

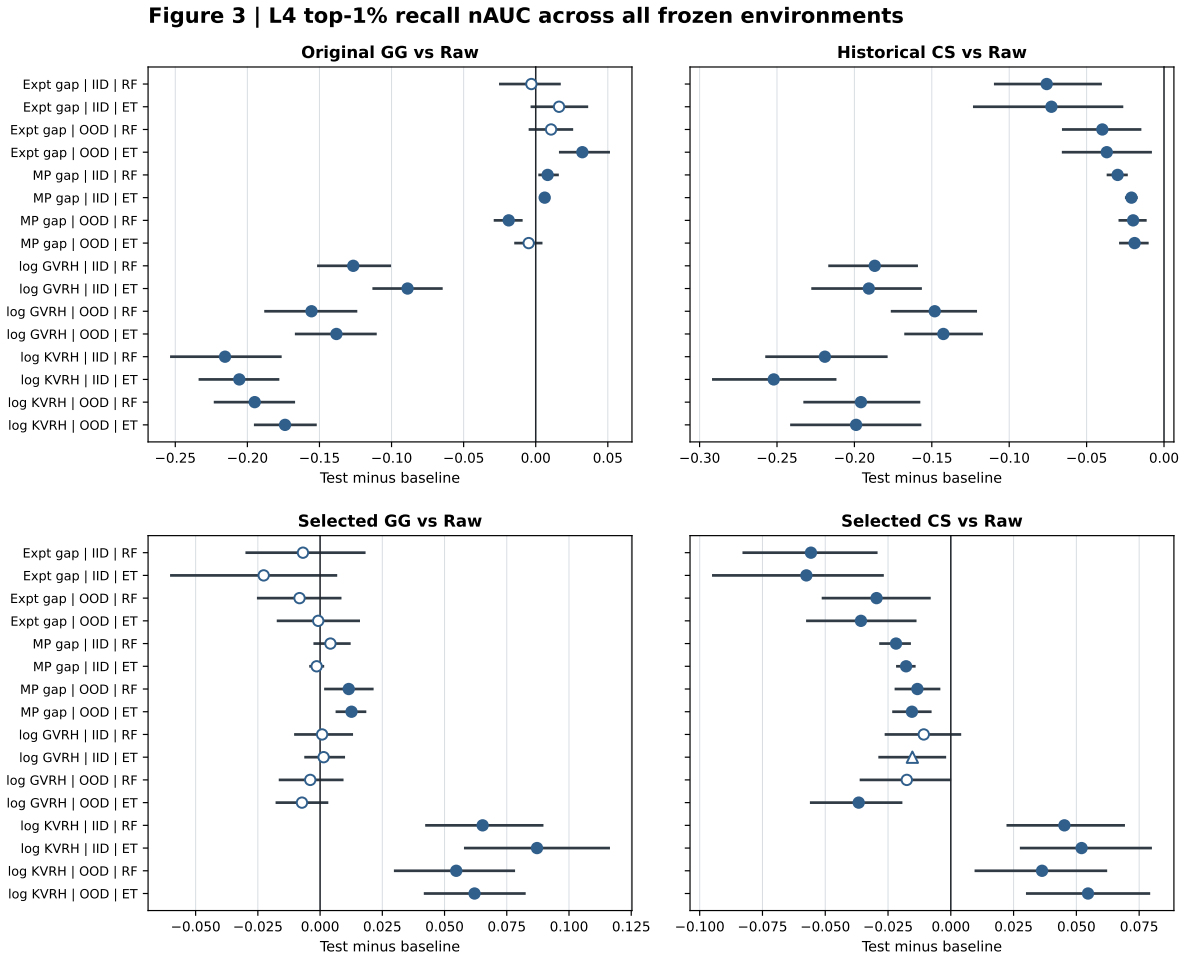


Figure S3: Historical/development evidence – not part of V6 confirmatory inference. V5 L4 environment-level effects.

Figure 4 | Original and Selected GG mechanism diagnostics

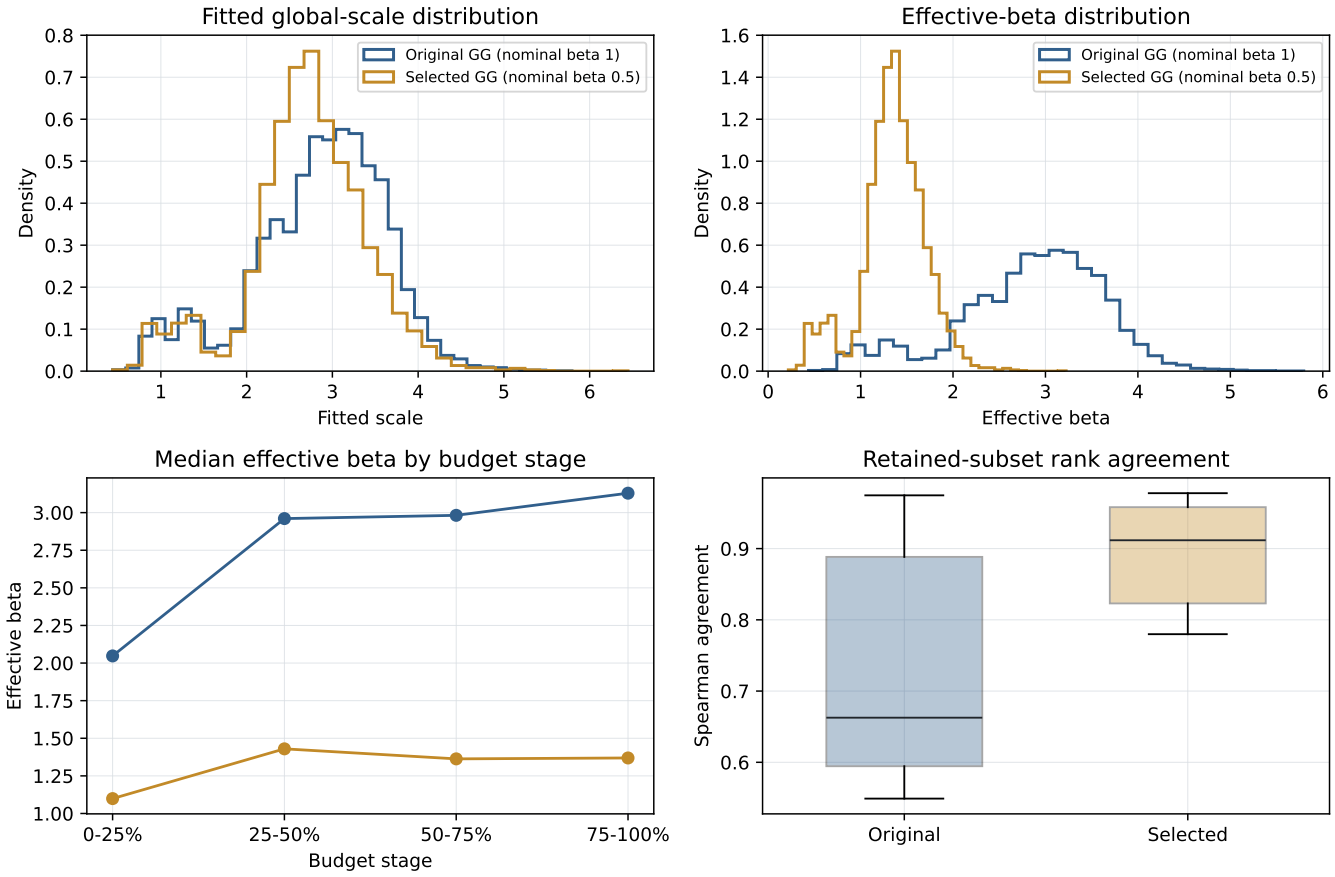


Figure S4: Historical/development evidence – not part of V6 confirmatory inference. Exploratory V5 GG diagnostics; no causal mechanism or full-pool rank equivalence is established.

S3 Complete registered pooled contrasts

Table S2: Complete locked V6 pooled registered contrasts. Effects are strategy minus baseline.

ID	Endpoint	Strategy	Baseline	Effect	CI low	CI high	Raw p	BH p	Same sign
H1	L1	GG_CALIBRATION_ONLY	RAW	0.0657770225	0.0627818431	0.0689712498	1	1	16/16
H2A	L4	GG_ACQUISITION_ONLY	RAW	-0.0763127258	-0.0852825362	-0.0676855935	1	1	11/16
H2B	L4	ORIGINAL_GG	RAW	-0.0763127258	-0.0853933246	-0.0676125952	1	1	11/16
H3	L4	RAW_FIXED_BETA_MATCHED	ORIGINAL_GG	-0.00023554277	-0.00599157738	0.00544599797	0.936106389	0.936106389	7/16
H4	L4	RAW_STAGE_BETA_MATCHED	ORIGINAL_GG	0.0316188586	0.0255350817	0.0376130635	9.9990001e-05	0.000199980002	16/16
H5A	L1	CS_CALIBRATION_ONLY	RAW	0.0958984563	0.0908273397	0.100887545	1	1	8/16
H5B	L4	CS_ACQUISITION_ONLY	RAW	-0.0340839368	-0.039446183	-0.0285985897	1	1	16/16
H5C	L4	SELECTED_CS	RAW	-0.0340839368	-0.0396059304	-0.028612379	1	1	16/16
H6	L1	SCALAR_TEMPERATURE_RESIDUAL_RATIO	SELECTED_CS	-0.111526544	-0.116234863	-0.106875745	—	—	16/16
H6A	L1	SCALAR_TEMPERATURE_NLL	IDENTITY_CONTROL	0.0611615712	0.0569630348	0.06534071	1	1	11/16
H6B	L1	SCALAR_TEMPERATURE_RESIDUAL_RATIO	IDENTITY_CONTROL	-0.0214418429	-0.0236669741	-0.0191473043	9.9990001e-05	0.000199980002	8/16

Table S3: Absolute endpoint levels reconstructed directly from the locked seed-level endpoint table. Absolute endpoint levels are shown for interpretation only. No additional inferential test was introduced.

Contrast	Baseline pooled mean	Strategy pooled mean	Registered effect
Stagewise matched-beta vs Original GG	0.565322	0.596941	0.031619
Residual-ratio scaling vs Identity Control	0.069511	0.048069	-0.021442
GG acquisition-only vs RAW	0.641635	0.565322	-0.076313
CS acquisition-only vs RAW	0.641635	0.607551	-0.034084

S4 Multiplicity families

Table S4: Exact frozen Benjamini–Hochberg family membership and decisions.

Family	Size	ID	Raw p	BH rank	Adjusted p	Reject
P0_L1_CALIBRATION_ONLY	2	H1	1	1	1	False
P0_L1_CALIBRATION_ONLY	2	H5A	1	2	1	False
P0_L1_SIMPLE_SCALING	2	H6B	9.9990001e-05	1	0.000199980002	True
P0_L1_SIMPLE_SCALING	2	H6A	1	2	1	False
P0_L4_ACQUISITION_AND_JOINT	4	H2A	1	1	1	False
P0_L4_ACQUISITION_AND_JOINT	4	H2B	1	2	1	False
P0_L4_ACQUISITION_AND_JOINT	4	H5B	1	3	1	False
P0_L4_ACQUISITION_AND_JOINT	4	H5C	1	4	1	False
P0_L4_MATCHED_BETA_MECHANISM	2	H4	9.9990001e-05	1	0.000199980002	True
P0_L4_MATCHED_BETA_MECHANISM	2	H3	0.936106389	2	0.936106389	False

S5 Task-, split-, model-, and environment-level diagnostics

Table S5: All locked V6 heterogeneity-diagnostic estimates. These rows do not replace pooled inference.

ID	Level	Stratum	n	Effect	CI low	CI high
H1	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.084107	0.077711	0.090402
H1	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.078466	0.070186	0.086129
H1	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.080554	0.072608	0.087660
H1	environment	matbench_expt_gap__iid_record__random_forest	40	0.075068	0.067445	0.082234
H1	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	0.054258	0.045882	0.062939
H1	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	0.045109	0.036312	0.054102
H1	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	0.036115	0.025907	0.045888
H1	environment	matbench_log_gvrh__iid_record__random_forest	40	0.032798	0.022670	0.042386
H1	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	0.036649	0.025782	0.046830
H1	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	0.027242	0.015809	0.038070
H1	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	0.006948	-0.002153	0.016197
H1	environment	matbench_log_kvrh__iid_record__random_forest	40	0.004556	-0.004279	0.013493
H1	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.129270	0.128296	0.130231
H1	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	0.126928	0.125594	0.128185
H1	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.119730	0.118852	0.120634
H1	environment	matbench_mp_gap__iid_record__random_forest	40	0.114635	0.113597	0.115633
H1	model	extra_randomized_bagging	40	0.068454	0.065258	0.071724
H1	model	random_forest	40	0.063100	0.059816	0.066335
H1	split	chemical_system_ood	40	0.072754	0.068469	0.076860
H1	split	iid_record	40	0.058800	0.054330	0.063047
H1	task	matbench_expt_gap	40	0.079549	0.074796	0.084130
H1	task	matbench_log_gvrh	40	0.042070	0.035803	0.048462
H1	task	matbench_log_kvrh	40	0.018849	0.011436	0.026304
H1	task	matbench_mp_gap	40	0.122641	0.121946	0.123327
H2A	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.002716	-0.014063	0.022992
H2A	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.019122	-0.001340	0.044569
H2A	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.038690	0.019196	0.059822
H2A	environment	matbench_expt_gap__iid_record__random_forest	40	0.056585	0.029091	0.087094
H2A	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.124075	-0.152186	-0.097816
H2A	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.154710	-0.194730	-0.114831
H2A	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.102748	-0.128515	-0.078263
H2A	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.115820	-0.147403	-0.086942
H2A	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.190525	-0.222846	-0.157921
H2A	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.203504	-0.243250	-0.164458
H2A	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.207599	-0.238918	-0.176780
H2A	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.223114	-0.252158	-0.194854
H2A	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	-0.008352	-0.016352	-0.000369
H2A	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.016904	-0.029402	-0.004149
H2A	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.009239	0.006016	0.012549
H2A	environment	matbench_mp_gap__iid_record__random_forest	40	-4.312087e-06	-0.004693	0.005413
H2A	model	extra_randomized_bagging	40	-0.072832	-0.081759	-0.064022
H2A	model	random_forest	40	-0.079794	-0.091354	-0.068738
H2A	split	chemical_system_ood	40	-0.084529	-0.097204	-0.072333
H2A	split	iid_record	40	-0.068096	-0.077664	-0.058561
H2A	task	matbench_expt_gap	40	0.029278	0.017290	0.041686
H2A	task	matbench_log_gvrh	40	-0.124338	-0.144592	-0.105010
H2A	task	matbench_log_kvrh	40	-0.206186	-0.226131	-0.186307
H2A	task	matbench_mp_gap	40	-0.004005	-0.008536	0.000821
H2B	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.002716	-0.014025	0.022991
H2B	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.019122	-0.001376	0.043900
H2B	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.038690	0.019271	0.060528
H2B	environment	matbench_expt_gap__iid_record__random_forest	40	0.056585	0.028682	0.087613
H2B	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.124075	-0.151816	-0.097151
H2B	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.154710	-0.192960	-0.114391
H2B	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.102748	-0.128747	-0.078516
H2B	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.115820	-0.146944	-0.086554
H2B	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.190525	-0.222772	-0.158726
H2B	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.203504	-0.243158	-0.164391

ID	Level	Stratum	n	Effect	CI low	CI high
H2B	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.207599	-0.239174	-0.175949
H2B	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.223114	-0.251640	-0.193652
H2B	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	-0.008352	-0.016532	-0.000467
H2B	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.016904	-0.029572	-0.004084
H2B	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.009239	0.006144	0.012624
H2B	environment	matbench_mp_gap__iid_record__random_forest	40	-4.312087e-06	-0.004738	0.005516
H2B	model	extra_randomized_bagging	40	-0.072832	-0.081864	-0.064070
H2B	model	random_forest	40	-0.079794	-0.091015	-0.068594
H2B	split	chemical_system_ood	40	-0.084529	-0.097308	-0.072548
H2B	split	iid_record	40	-0.068096	-0.077526	-0.058714
H2B	task	matbench_expt_gap	40	0.029278	0.017493	0.041984
H2B	task	matbench_log_gvrh	40	-0.124338	-0.144975	-0.105000
H2B	task	matbench_log_kvrh	40	-0.206186	-0.225986	-0.186328
H2B	task	matbench_mp_gap	40	-0.004005	-0.008589	0.000792
H3	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.011384	-0.008891	0.032366
H3	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	-0.009487	-0.03125	0.013021
H3	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.038616	0.009523	0.068899
H3	environment	matbench_expt_gap__iid_record__random_forest	40	0.016964	-0.009822	0.044792
H3	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	0.018736	-0.017116	0.054253
H3	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	0.015754	-0.026487	0.059932
H3	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.038591	-0.064313	-0.013540
H3	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.023146	-0.047173	0.000408
H3	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.018330	-0.045630	0.009419
H3	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	0.021579	-0.005177	0.047827
H3	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.038727	-0.066887	-0.011150
H3	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.014210	-0.045742	0.015501
H3	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.010080	0.001029	0.019336
H3	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.001637	-0.016351	0.012246
H3	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.000319	-0.004208	0.004515
H3	environment	matbench_mp_gap__iid_record__random_forest	40	0.006926	0.001380	0.012511
H3	model	extra_randomized_bagging	40	-0.002064	-0.010785	0.006454
H3	model	random_forest	40	0.001593	-0.006083	0.009201
H3	split	chemical_system_ood	40	0.006010	-0.002261	0.013818
H3	split	iid_record	40	-0.006481	-0.015996	0.002647
H3	task	matbench_expt_gap	40	0.014369	0.002018	0.027111
H3	task	matbench_log_gvrh	40	-0.006812	-0.024133	0.010580
H3	task	matbench_log_kvrh	40	-0.012422	-0.023609	-0.001090
H3	task	matbench_mp_gap	40	0.003922	-0.000685	0.008631
H4	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.011793	-0.006027	0.031027
H4	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.009263	-0.012353	0.030990
H4	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.007403	-0.019159	0.033222
H4	environment	matbench_expt_gap__iid_record__random_forest	40	0.005841	-0.017970	0.030506
H4	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	0.061412	0.030933	0.093821
H4	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	0.093914	0.057222	0.132736
H4	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	0.043077	0.022701	0.064131
H4	environment	matbench_log_gvrh__iid_record__random_forest	40	0.047956	0.028772	0.067214
H4	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	0.043545	0.014779	0.071399
H4	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	0.052052	0.017550	0.087153
H4	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	0.030632	0.007122	0.054725
H4	environment	matbench_log_kvrh__iid_record__random_forest	40	0.047268	0.021556	0.071626
H4	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.020238	0.011354	0.028547
H4	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	0.014976	0.002968	0.026551
H4	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.003689	0.000528	0.006776
H4	environment	matbench_mp_gap__iid_record__random_forest	40	0.012842	0.009068	0.016453
H4	model	extra_randomized_bagging	40	0.027724	0.019011	0.036538
H4	model	random_forest	40	0.035514	0.026546	0.044608
H4	split	chemical_system_ood	40	0.038399	0.029340	0.047501
H4	split	iid_record	40	0.024839	0.018394	0.031503
H4	task	matbench_expt_gap	40	0.008575	-0.002158	0.019811
H4	task	matbench_log_gvrh	40	0.061590	0.045212	0.078379
H4	task	matbench_log_kvrh	40	0.043374	0.027517	0.059380
H4	task	matbench_mp_gap	40	0.012936	0.009464	0.016392
H5A	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.212830	0.175166	0.250399
H5A	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.287251	0.263011	0.310478
H5A	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.155433	0.125889	0.185603

ID	Level	Stratum	n	Effect	CI low	CI high
H5A	environment	matbench_expt_gap__iid_record__random_forest	40	0.270989	0.244927	0.295372
H5A	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.013605	-0.020555	-0.007206
H5A	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.021731	-0.027960	-0.015302
H5A	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.017066	-0.022430	-0.011778
H5A	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.019095	-0.024673	-0.013404
H5A	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.014287	-0.020605	-0.008202
H5A	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.019810	-0.026138	-0.013425
H5A	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.019255	-0.025694	-0.012638
H5A	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.021493	-0.027168	-0.015513
H5A	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.198331	0.191248	0.205578
H5A	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	0.215927	0.209543	0.221960
H5A	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.157429	0.151777	0.163066
H5A	environment	matbench_mp_gap__iid_record__random_forest	40	0.182527	0.176240	0.188935
H5A	model	extra_randomized_bagging	40	0.082476	0.075551	0.089331
H5A	model	random_forest	40	0.109321	0.104389	0.114167
H5A	split	chemical_system_ood	40	0.105613	0.098401	0.112813
H5A	split	iid_record	40	0.086184	0.080201	0.091986
H5A	task	matbench_expt_gap	40	0.231626	0.212818	0.250477
H5A	task	matbench_log_gvrh	40	-0.017874	-0.022122	-0.013716
H5A	task	matbench_log_kvrh	40	-0.018711	-0.022840	-0.014481
H5A	task	matbench_mp_gap	40	0.188553	0.184313	0.192588
H5B	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	-0.056436	-0.079912	-0.033854
H5B	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	-0.038876	-0.063207	-0.017856
H5B	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	-0.058296	-0.079391	-0.038057
H5B	environment	matbench_expt_gap__iid_record__random_forest	40	-0.017671	-0.047136	0.017523
H5B	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.027706	-0.044001	-0.011738
H5B	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.017409	-0.037839	0.004431
H5B	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.028129	-0.042291	-0.014085
H5B	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.028916	-0.046794	-0.010460
H5B	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.044466	-0.060395	-0.028997
H5B	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.031536	-0.050604	-0.013656
H5B	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.042027	-0.063064	-0.020632
H5B	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.073210	-0.102535	-0.047653
H5B	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	-0.017617	-0.026321	-0.008136
H5B	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.027299	-0.037352	-0.017764
H5B	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	-0.013215	-0.015894	-0.010495
H5B	environment	matbench_mp_gap__iid_record__random_forest	40	-0.022531	-0.027667	-0.017620
H5B	model	extra_randomized_bagging	40	-0.035987	-0.042466	-0.029531
H5B	model	random_forest	40	-0.032181	-0.039387	-0.025273
H5B	split	chemical_system_ood	40	-0.032668	-0.039989	-0.025586
H5B	split	iid_record	40	-0.035500	-0.043571	-0.027042
H5B	task	matbench_expt_gap	40	-0.042820	-0.056828	-0.029539
H5B	task	matbench_log_gvrh	40	-0.025540	-0.035461	-0.015850
H5B	task	matbench_log_kvrh	40	-0.047810	-0.058794	-0.037148
H5B	task	matbench_mp_gap	40	-0.020166	-0.023445	-0.017122
H5C	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	-0.056436	-0.080953	-0.033853
H5C	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	-0.038876	-0.062463	-0.017783
H5C	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	-0.058296	-0.079577	-0.038504
H5C	environment	matbench_expt_gap__iid_record__random_forest	40	-0.017671	-0.047321	0.016481
H5C	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.027706	-0.043655	-0.011714
H5C	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.017409	-0.037566	0.003898
H5C	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.028129	-0.042282	-0.013853
H5C	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.028916	-0.046888	-0.009748
H5C	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.044466	-0.060395	-0.028908
H5C	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.031536	-0.050735	-0.013133
H5C	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.042027	-0.063710	-0.021499
H5C	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.073210	-0.102117	-0.048039
H5C	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	-0.017617	-0.026495	-0.008467
H5C	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.027299	-0.037282	-0.017603
H5C	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	-0.013215	-0.015907	-0.010406
H5C	environment	matbench_mp_gap__iid_record__random_forest	40	-0.022531	-0.027772	-0.017678
H5C	model	extra_randomized_bagging	40	-0.035987	-0.042494	-0.029724
H5C	model	random_forest	40	-0.032181	-0.039246	-0.025210
H5C	split	chemical_system_ood	40	-0.032668	-0.040051	-0.025630
H5C	split	iid_record	40	-0.035500	-0.043497	-0.027011

ID	Level	Stratum	n	Effect	CI low	CI high
H5C	task	matbench_expt_gap	40	-0.042820	-0.057041	-0.029594
H5C	task	matbench_log_gvrh	40	-0.025540	-0.035104	-0.015893
H5C	task	matbench_log_kvrh	40	-0.047810	-0.059070	-0.036995
H5C	task	matbench_mp_gap	40	-0.020166	-0.023285	-0.017145
H6	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	-0.186395	-0.220548	-0.153343
H6	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	-0.257999	-0.283112	-0.232080
H6	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	-0.131703	-0.161756	-0.103738
H6	environment	matbench_expt_gap__iid_record__random_forest	40	-0.244785	-0.271043	-0.217652
H6	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.035359	-0.041567	-0.029168
H6	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.039671	-0.048136	-0.031360
H6	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.039010	-0.047604	-0.030665
H6	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.042502	-0.049716	-0.035405
H6	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.009900	-0.015757	-0.004175
H6	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.029103	-0.038255	-0.020573
H6	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.014470	-0.023780	-0.005275
H6	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.022656	-0.031471	-0.013979
H6	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	-0.181475	-0.187389	-0.175777
H6	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	-0.200570	-0.206590	-0.194593
H6	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	-0.164603	-0.170464	-0.158517
H6	environment	matbench_mp_gap__iid_record__random_forest	40	-0.184224	-0.190734	-0.177572
H6	model	extra_randomized_bagging	40	-0.095364	-0.101306	-0.089488
H6	model	random_forest	40	-0.127689	-0.132677	-0.122864
H6	split	chemical_system_ood	40	-0.117559	-0.124437	-0.110833
H6	split	iid_record	40	-0.105494	-0.112387	-0.098963
H6	task	matbench_expt_gap	40	-0.205220	-0.222402	-0.189137
H6	task	matbench_log_gvrh	40	-0.039135	-0.044170	-0.034100
H6	task	matbench_log_kvrh	40	-0.019032	-0.022978	-0.014970
H6	task	matbench_mp_gap	40	-0.182718	-0.186088	-0.179416
H6A	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.064865	0.053623	0.076302
H6A	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.082756	0.072421	0.092888
H6A	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.050073	0.037669	0.062577
H6A	environment	matbench_expt_gap__iid_record__random_forest	40	0.077001	0.063282	0.090158
H6A	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	0.022089	0.008750	0.035321
H6A	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	0.041248	0.029207	0.052658
H6A	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.000459	-0.014196	0.013765
H6A	environment	matbench_log_gvrh__iid_record__random_forest	40	0.008731	-0.006643	0.023556
H6A	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.010169	-0.024195	0.003113
H6A	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.006860	-0.020916	0.006951
H6A	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.021980	-0.034421	-0.009303
H6A	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.024223	-0.036686	-0.011607
H6A	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.174401	0.171531	0.176696
H6A	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	0.175055	0.173960	0.176081
H6A	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.169596	0.163580	0.174857
H6A	environment	matbench_mp_gap__iid_record__random_forest	40	0.176462	0.174960	0.177813
H6A	model	extra_randomized_bagging	40	0.056052	0.052002	0.060147
H6A	model	random_forest	40	0.066271	0.061663	0.070947
H6A	split	chemical_system_ood	40	0.067923	0.062326	0.073548
H6A	split	iid_record	40	0.054400	0.048110	0.060468
H6A	task	matbench_expt_gap	40	0.068674	0.060894	0.076053
H6A	task	matbench_log_gvrh	40	0.017902	0.009664	0.026062
H6A	task	matbench_log_kvrh	40	-0.015808	-0.026226	-0.005775
H6A	task	matbench_mp_gap	40	0.173878	0.172161	0.175465
H6B	environment	matbench_expt_gap__chemical_system_ood__extra_randomized_bagging	40	0.005357	0.002355	0.008413
H6B	environment	matbench_expt_gap__chemical_system_ood__random_forest	40	0.010595	0.006963	0.014312
H6B	environment	matbench_expt_gap__iid_record__extra_randomized_bagging	40	0.002236	-0.002907	0.007159
H6B	environment	matbench_expt_gap__iid_record__random_forest	40	0.004162	0.000673	0.007817
H6B	environment	matbench_log_gvrh__chemical_system_ood__extra_randomized_bagging	40	-0.051963	-0.059725	-0.044084
H6B	environment	matbench_log_gvrh__chemical_system_ood__random_forest	40	-0.061209	-0.069459	-0.052739
H6B	environment	matbench_log_gvrh__iid_record__extra_randomized_bagging	40	-0.055871	-0.062968	-0.048615
H6B	environment	matbench_log_gvrh__iid_record__random_forest	40	-0.064399	-0.072030	-0.056621
H6B	environment	matbench_log_kvrh__chemical_system_ood__extra_randomized_bagging	40	-0.032768	-0.038305	-0.027097
H6B	environment	matbench_log_kvrh__chemical_system_ood__random_forest	40	-0.045664	-0.053386	-0.038384
H6B	environment	matbench_log_kvrh__iid_record__extra_randomized_bagging	40	-0.043414	-0.049365	-0.037359
H6B	environment	matbench_log_kvrh__iid_record__random_forest	40	-0.052426	-0.058631	-0.046067
H6B	environment	matbench_mp_gap__chemical_system_ood__extra_randomized_bagging	40	0.019515	0.017838	0.021256

ID	Level	Stratum	n	Effect	CI low	CI high
H6B	environment	matbench_mp_gap__chemical_system_ood__random_forest	40	0.018226	0.016566	0.019967
H6B	environment	matbench_mp_gap__iid_record__extra_randomized_bagging	40	0.002132	0.001394	0.002884
H6B	environment	matbench_mp_gap__iid_record__random_forest	40	0.002420	0.001734	0.003103
H6B	model	extra_randomized_bagging	40	-0.019347	-0.021617	-0.017096
H6B	model	random_forest	40	-0.023537	-0.026038	-0.020982
H6B	split	chemical_system_ood	40	-0.017239	-0.020071	-0.014336
H6B	split	iid_record	40	-0.025645	-0.028717	-0.022584
H6B	task	matbench_expt_gap	40	0.005588	0.003188	0.007843
H6B	task	matbench_log_gvrh	40	-0.058360	-0.063702	-0.052623
H6B	task	matbench_log_kvrh	40	-0.043568	-0.047583	-0.039520
H6B	task	matbench_mp_gap	40	0.010573	0.009861	0.011319

S6 Pair completeness

Table S6: All 176 registered environment-level completeness gates.

ID	Task	Split	Model	Pairs	Missing	Excl.	Gate
H1	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H1	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H1	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H1	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H1	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H1	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H1	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H1	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H1	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H1	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H1	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H1	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H1	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H1	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H1	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H1	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H2A	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H2A	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H2A	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H2A	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H2A	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H2A	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H2A	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H2A	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H2A	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H2B	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H2B	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H2B	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H2B	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H2B	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H2B	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H2B	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H2B	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H2B	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H3	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H3	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H3	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H3	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H3	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H3	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H3	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H3	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H3	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H3	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS

ID	Task	Split	Model	Pairs	Missing	Excl.	Gate
H3	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H3	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H3	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H3	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H3	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H3	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H4	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H4	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H4	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H4	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H4	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H4	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H4	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H4	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H4	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H4	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H4	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H4	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H4	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H4	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H4	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H4	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H5A	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5A	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H5A	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5A	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H5A	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5A	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H5A	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5A	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5A	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H5B	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5B	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H5B	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5B	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H5B	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5B	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H5B	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5B	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5B	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H5C	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5C	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H5C	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5C	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H5C	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H5C	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS

ID	Task	Split	Model	Pairs	Missing	Excl.	Gate
H5C	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H5C	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H5C	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H5C	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H6	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H6	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H6	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H6	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H6A	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6A	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H6A	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6A	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H6A	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6A	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H6A	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6A	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6A	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS
H6B	matbench_expt_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_expt_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6B	matbench_expt_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_expt_gap	iid_record	random_forest	40	0	0	PASS
H6B	matbench_log_gvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_log_gvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6B	matbench_log_gvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_log_gvrh	iid_record	random_forest	40	0	0	PASS
H6B	matbench_log_kvrh	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_log_kvrh	chemical_system_ood	random_forest	40	0	0	PASS
H6B	matbench_log_kvrh	iid_record	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_log_kvrh	iid_record	random_forest	40	0	0	PASS
H6B	matbench_mp_gap	chemical_system_ood	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_mp_gap	chemical_system_ood	random_forest	40	0	0	PASS
H6B	matbench_mp_gap	iid_record	extra_randomized_bagging	40	0	0	PASS
H6B	matbench_mp_gap	iid_record	random_forest	40	0	0	PASS

S7 H7 descriptive direction table

Table S7: Complete H7 direction table. H7 is descriptive and generates no binary claim.

H7 row	Source	Endpoint	Strategy	Baseline	Effect	Direction	Binary
H7__H1	H1	L1	GG_CALIBRATION_ONLY	RAW	0.065777	unfavorable	False
H7__H5A	H5A	L1	CS_CALIBRATION_ONLY	RAW	0.095898	unfavorable	False
H7__H2A	H2A	L4	GG_ACQUISITION_ONLY	RAW	-0.076313	unfavorable	False
H7__H2B	H2B	L4	ORIGINAL_GG	RAW	-0.076313	unfavorable	False
H7__H5B	H5B	L4	CS_ACQUISITION_ONLY	RAW	-0.034084	unfavorable	False
H7__H5C	H5C	L4	SELECTED_CS	RAW	-0.034084	unfavorable	False
H7__H3	H3	L4	RAW_FIXED_BETA_MATCHED	ORIGINAL_GG	-0.000236	unfavorable	False
H7__H4	H4	L4	RAW_STAGE_BETA_MATCHED	ORIGINAL_GG	0.031619	favorable	False
H7__H6A	H6A	L1	SCALAR_TEMPERATURE_NLL	IDENTITY_CONTROL	0.061162	unfavorable	False
H7__H6B	H6B	L1	SCALAR_TEMPERATURE_RESIDUAL_RATIO	IDENTITY_CONTROL	-0.021442	favorable	False
H7__H6	H6	L1	SCALAR_TEMPERATURE_RESIDUAL_RATIO	SELECTED_CS	-0.111527	favorable	False

S8 Strategy and endpoint registry

Table S8: Frozen endpoint registry.

ID	Name	Role	Direction	Unit
L1	campaign MACE at initial/25/50/75% canonical checkpoints	primary	lower_is_better	paired seed within environment
L4	top-1% recall normalized AUC	primary	higher_is_better	paired seed within environment
CONFORMAL_COVERAGE	coverage-specific conformal interval coverage	secondary_calibration	absolute error lower is better	
NLL	Gaussian negative log likelihood	secondary_conditional	lower_is_better	
RANKING_DIAGNOSTICS	stable-order path diagnostics	mechanism_secondary	descriptive	

S8.1 Strategy display names and policy identity

Table S9: Frozen strategy, beta, mapping, and policy identities for the registered comparisons and controls.

Strategy	Policy key	Nominal base β	Calibration	Acquisition	Evaluation	Objective
IDENTITY_CONTROL	IDENTITY_CONTROL	1.0	raw_gaussian_sigma	raw_sigma	raw_gaussian_sigma	identity_control
ORIGINAL_GG	ORIGINAL_GG__beta_1	1.0	scaled_gaussian_sigma	scaled_sigma	scaled_gaussian_sigma	V5 Gaussian-consistent median residual-ratio
RAW_FIXED_BETA_MATCHED	RAW_FIXED_BETA_MATCHED	1.0	raw_gaussian_sigma	raw_sigma	raw_gaussian_sigma	fixed development-matched-beta
RAW_STAGE_BETA_MATCHED	RAW_STAGE_BETA_MATCHED	1.0	raw_gaussian_sigma	raw_sigma	raw_gaussian_sigma	stagewise development-matched-beta
SCALAR_TEMPERATURE_NLL	SCALAR_TEMPERATURE_NLL	1.0	scaled_gaussian_sigma	raw_sigma	scaled_gaussian_sigma	Gaussian OOB NLL; calibration-only simple scaler
SCALAR_TEMPERATURE_RESIDUAL_RATIO	SCALAR_TEMPERATURE_RESIDUAL_RATIO	1.0	scaled_gaussian_sigma	raw_sigma	scaled_gaussian_sigma	OOB median absolute residual ratio; calibration-only simple scaler
SELECTED_CS	SELECTED_CS	1.0	scaled_gaussian_sigma	scaled_sigma	scaled_gaussian_sigma	V5 parsimonious F1 pairwise candidate-specific ranking
SELECTED_GG	SELECTED_GG__beta_0p5	0.5	scaled_gaussian_sigma	scaled_sigma	scaled_gaussian_sigma	V5 Gaussian-global scaler with development-selected beta

The operational exploration coefficient is β for RAW/identity, βc_t for GG, $\beta c_t(x)$ for CS, β_{fixed} for fixed matched-beta, and $\beta_{s(t)}$ for stagewise matched-beta. Display names are mapped to unchanged machine identifiers in `tables/display_name_mapping.csv`.

S8.2 Main Figure 1 strategy-role and source information

Main Figure 1 is an explanatory schematic and contains no effect estimate. Its policy roles were read from `analysis_plan/v6_p0_confirmation_selection_map_FROZEN.json`; endpoint roles were read from the frozen endpoint registry. The complete element-level binding, including source hashes and exact source rows, is in `tables/revised_figure_source_inventory.csv`. High-resolution outputs are supplied as `Figure1_uncertainty_role_decomposition.png`, `Figure1_uncertainty_role_decomposition.pdf`, and `Figure1_uncertainty_role_decomposition.svg`.

Role	Acquisition	L1 evaluation
Calibration-only	raw σ	transformed σ
Acquisition-only	transformed σ	raw σ
Joint	transformed σ	transformed σ
Fixed matched-beta	frozen constant beta	raw σ
Stagewise matched-beta	frozen beta schedule	raw σ
RAW/Identity	raw σ	raw σ
Simple scalar	raw σ	transformed σ

Strategy-role mapping used in Main Figure 1. Simple scalar methods enter calibration replay unless explicitly registered otherwise.

S9 Matched-beta coefficients

The columns labelled Fixed and by budget stage are the Operational exploration coefficient. The task-specific coefficients below were derived only from broad leave-one-task-out development seeds 6100–6109. For each held-out task, that task did not enter its own beta map. Confirmation seeds 7000–7039 were not used. The registered nominal beta was 1.0, so the fixed and stagewise values were the operational acquisition coefficients. Every value is bound to the exact frozen selection-map file SHA-256 printed in its row.

Table S10: Task-specific frozen operational matched-beta coefficients. The registered Nominal base β was 1.0; displayed values are the operational exploration coefficients.

Held-out task	Fixed	0–25%	25–50%	50–75%	75–100%	Frozen selection-map SHA-256
matbench_expt_gap	3.129251	1.886456	3.028321	3.242677	3.335345	77960873 3b1b7e87 27a28b3d 08666227 d6924175 6e30f685 180ea2ca 0f8909dd
matbench_log_gvrh	2.935900	1.816189	2.880623	2.983102	2.928673	77960873 3b1b7e87 27a28b3d 08666227 d6924175 6e30f685 180ea2ca 0f8909dd
matbench_log_kvrh	2.934119	1.851547	3.015790	2.952755	2.896339	77960873 3b1b7e87 27a28b3d 08666227 d6924175 6e30f685 180ea2ca 0f8909dd
matbench_mp_gap	2.996273	1.746245	2.880623	3.054085	3.080713	77960873 3b1b7e87 27a28b3d 08666227 d6924175 6e30f685 180ea2ca 0f8909dd

S10 Execution and result-lock provenance

Sentinel implementation validation used seeds 6000–6009. Broad leave-one-task-out development and policy selection used seeds 6100–6109, whereas confirmation used the disjoint seeds 7000–7039. The operational matched-beta maps above are bound to `779608733b1b7e8727a28b3d08666227d69241756e30f685180ea2ca0f8909dd`. All 12,800 frozen confirmation trajectory and replay jobs completed, all 176 completeness gates passed, and no endpoint was missing. The frozen inference lock records 11 primary contrast rows, 10 multiplicity rows, 264 heterogeneity rows, and 11 H7 rows.

Table S11: Files bound by the frozen V6 inference-result lock.

Locked object	Path	SHA-256
endpoint_table	tables/v6_p0_confirmation_seed_level_endpoints.csv	a56a8530 83535b76 b8e2c7ea eab056bd d81b0a9e 73e953fb 00c13237 c9c8af8d
endpoint_table_lock	locks/v6_p0_confirmation_seed_level_endpoints_lock.json	0ac56f85 4aadfe73 f488abb5 c33daa82 92246bb6 96dfe73b ceeba6ae 48f93d83
environment_effects	tables/v6_p0_confirmation_environment_effects.csv	379555bc 225bf8e0 52ce5aa2 01a52ee2 266d7b66 99a4e7b9 e59e699e e2e284e9
h7_direction_table	tables/v6_p0_confirmation_h7_direction_table.csv	dc2e75bf d0a867b6 e0681022 83e87961 a358f867 c5066200 fba77373 a9ee31bf
inference_report	reports/09_p0_confirmation_inference.md	084e72ad 09f2f96c ff709c6f 444af02b d40a5fcc b7182bda ec90259d 74c6d620
multiplicity	tables/v6_p0_confirmation_multiplicity.csv	f117dfaf 7766e7dc 648088c0 1d0e7ce1 e6a16b18 68c10959 4e77a69e fec20e56
pair_completeness	tables/v6_p0_confirmation_pair_completeness.csv	b6e8cf31 b19b2c43 19517192 4e292bf8 ab36acbb dbc4821c 72db8849 db6a7ce8
primary_contrasts	tables/v6_p0_confirmation_primary_contrasts.csv	2b9820a0 167a8e87 5c322d10 5fe54cdb ad614a39 1f02db59 938aabfe 06d2a2fa
raw_lock_validation_report	reports/08_p0_raw_result_lock_validation.md	75130637 47c62061 541ad97a de42e503 2be2d29f 0f84e351 60941b15 b6a5c6a8
resampling_seeds	tables/v6_p0_confirmation_resampling_seeds.csv	398f2528 4ddf6785 9f1afed8 d51d7d6a 8e328ac5 ec3224c7 bc0d85ef 2a9aa6ee

S11 Related-work comparison

Table S12 positions the operational contribution against studies whose primary records and identifiers were verified. Different endpoints and intervention structures are not treated as replications of the present design.

Table S12: Verified related work and the role-decomposed contribution.

Study	Calibration endpoint	Discovery endpoint	Surrogate/domain	Role separation	Independent confirmation	Difference from present work
Foldager et al. (2023), PMLR 216:592-601; arXiv:2301.05983	Regression uncertainty calibration	Bayesian-optimization regret	Synthetic and real-world BO; multiple surrogates	No intervention-level calibration/acquisition decomposition	Benchmark repetitions	The present study freezes role-specific interventions and separates calibration from discovery endpoints.
Wang et al. (2022), doi:10.1093/oxfmat/itac006	Not a primary endpoint	Materials optimization regret	Fe-Co-Ni experimental library; Gaussian process	No	Repeated benchmark campaigns	The present study adds locked calibration endpoints and role-specific controls.
Varivoda et al. (2023), doi:10.1063/5.0133528	UQ quality and error bounds	Active-learning utility	Materials property prediction; graph models	Uses compared, not intervention roles	Not reported as a frozen confirmation stage	The present study separates evaluation and acquisition mappings within one campaign architecture.
Jiang et al. (2026), doi:10.20517/jmi.2025.70	Miscalibration area	Optimization across alloy datasets	SVR, neural network and XGBoost; three alloy datasets	No calibration-only versus acquisition-only arms	No frozen paired confirmation stage reported	The present study tests whether scaling transfers across roles and surrogate constructions.
Dale et al. (2025 preprint), arXiv:2511.17760	ID/OOD uncertainty calibration	OOD active-learning data efficiency	ALIGNN, XGBoost, Random Forest and neural-network ensembles	Calibration methods evaluated through acquisition	Five active-learning seeds; no frozen confirmation stage reported	The present study uses locked paired confirmation and separates evaluation from acquisition placement.
Riebesell et al. (2025), doi:10.1038/s42256-025-01055-1	Reliability and error diagnostics	Stability classification and discovery acceleration	Crystal-stability prefilter benchmark	No	Prospective benchmark framing	The present study adds frozen role-specific contrasts and matched-beta controls.
Present study	Four-coverage campaign MACE	Top-1% recall nAUC	Four Matbench task snapshots; tree and QGB surrogates	Calibration-only and acquisition-only interventions	Forty disjoint paired confirmation seeds	Frozen role-specific comparisons with matched-exploration controls

S12 P1 protocol and evidence boundary

P1 used disjoint sentinel (8000–8004), development (8100–8109) and confirmation (8200–8239) seeds. Five contrasts, two multiplicity families, endpoints, directions, exclusions and equal-environment paired-seed aggregation were frozen before confirmation. A compute-only amendment, created at 2026-08-01T18:30:00+08:00 before any confirmation outcome, changed only wall-time, worker and scheduling limits. No P1 value entered V6 inference.

S13 P1 surrogate and strategy registry

The non-bagged surrogate comprised one squared-error and two quantile gradient-boosting regressors at 0.16 and 0.84. The quantile width defines a conditional predictive-spread proxy with a 10^{-8} floor, not a strict epistemic-uncertainty estimate.

Table S13: P1 strategy-role map.

Strategy	Surrogate	Acquisition uncertainty	Evaluation uncertainty	Scalar source	Registered endpoint role
QGB raw	quantile gradient boosting	raw quantile spread	raw quantile spread	none	P1_L1, P1_L4
QGB residual-ratio calibration-only	quantile gradient boosting	raw quantile spread	cross-fitted residual-ratio scaled quantile spread	labelled five-fold cross-fit residual ratio	P1_L1
QGB residual-ratio acquisition-only	quantile gradient boosting	cross-fitted residual-ratio scaled quantile spread	raw quantile spread	labelled five-fold cross-fit residual ratio	P1_L4
Tree RAW	frozen V6 bagged tree	raw tree dispersion	raw tree dispersion	none	P1_L4
Tree fixed matched beta	frozen V6 bagged tree	fixed matched-beta tree dispersion	raw tree dispersion	frozen V6 development map	P1_L4
Tree stagewise matched beta	frozen V6 bagged tree	stagewise matched-beta tree dispersion	raw tree dispersion	frozen V6 development map	P1_L4
Tree development-selected constant beta	frozen V6 bagged tree	development-selected constant-beta tree dispersion	raw tree dispersion	P1 development leave-one-task-out selection	P1_L4

S14 P1 complete registered contrasts

Table S14: Complete P1 registered comparisons and locked results.

ID	Strategy	Baseline	End.	Alt.	Family	Effect	95% CI	Raw p	BH p	Env.	Seeds	Decision
P1_C1	RAW_STAGE_BETA_MATCHED	RAW	L4	greater	P1_DISCOVERY_EXISTING	-0.040800	[-0.047582, -0.033996]	1.0000	1.0000	16	40	opposed registered alternative
P1_C2	RAW_STAGE_BETA_MATCHED	RAW_FIXED_BETA_MATCHED	L4	greater	P1_DISCOVERY_EXISTING	0.030320	[0.024001, 0.036682]	9.999×10^{-5}	0.00029997	16	40	supported
P1_C3	DEVELOPMENT_SELECTED_CONSTANT_BETA_UCB	RAW	L4	greater	P1_DISCOVERY_EXISTING	-0.017442	[-0.020936, -0.013972]	1.0000	1.0000	16	40	opposed registered alternative
P1_C4	P1_QGB_CALIBRATION_ONLY_RR	P1_QGB_RAW	L1	less	P1_NEW_UQ_ROLE	0.097439	[0.095013, 0.099841]	1.0000	1.0000	8	40	opposed registered alternative
P1_C5	P1_QGB_ACQUISITION_ONLY_RR	P1_QGB_RAW	L4	greater	P1_NEW_UQ_ROLE	-0.009354	[-0.014256, -0.004685]	1.0000	1.0000	8	40	opposed registered alternative

S15 P1 multiplicity families

The three tree L4 comparisons form one BH family; the two QGB role comparisons form a second. Neither shares adjustment with V6.

S16 P1 completeness

All 3,520 immutable-manifest jobs completed. Every registered pair contained all 40 confirmation seeds; duplicate jobs, evidence-stage contamination, environment selection and incomplete-subset inference were absent.

S17 P1 task, split and environment diagnostics

Figure S6 reports the four scientific tasks. Figures S20–S22 retain all task × split × model environments in readable groups. These rows are descriptive diagnostics rather than independent task replicates.

S18 P1 calibration diagnostics

Figures S8–S12 report coverage, sharpness, proper scores and residual diagnostics. Only campaign MACE in P1_C4 is confirmatory.

S19 P1 discovery diagnostics

Figures S13–S15 report recall, realized improvement, regret and budget-segment behavior. Only registered L4 comparisons are confirmatory.

S20 P1 robustness and sensitivity analyses

Task estimates, split summaries, seed-block distributions, leave-one-task-out estimates, between-task random effects and Wilcoxon sensitivity are exploratory robustness analyses.

S21 P1 execution provenance and result locks

Execution began at 2026-08-01T11:45:33.733115+00:00 and ended at 2026-08-01T20:51:14.238681+00:00, an observed elapsed wall time of 9.094585 h. Timestamp overlap shows at most six concurrent jobs on one audited host. The sum of job-process wall times was 53.650284 job-hours and is not labelled CPU-hours. The durable confirmation corpus occupied 68.592643 GiB after completion. No job failed and no retry marker was present. Peak RAM, CPU time and peak temporary-disk use were not instrumented and are not estimated retrospectively. Every job has an immutable manifest row, atomic state, execution log and provenance record. Final endpoint and inference artifacts are bound by SHA-256 result locks.

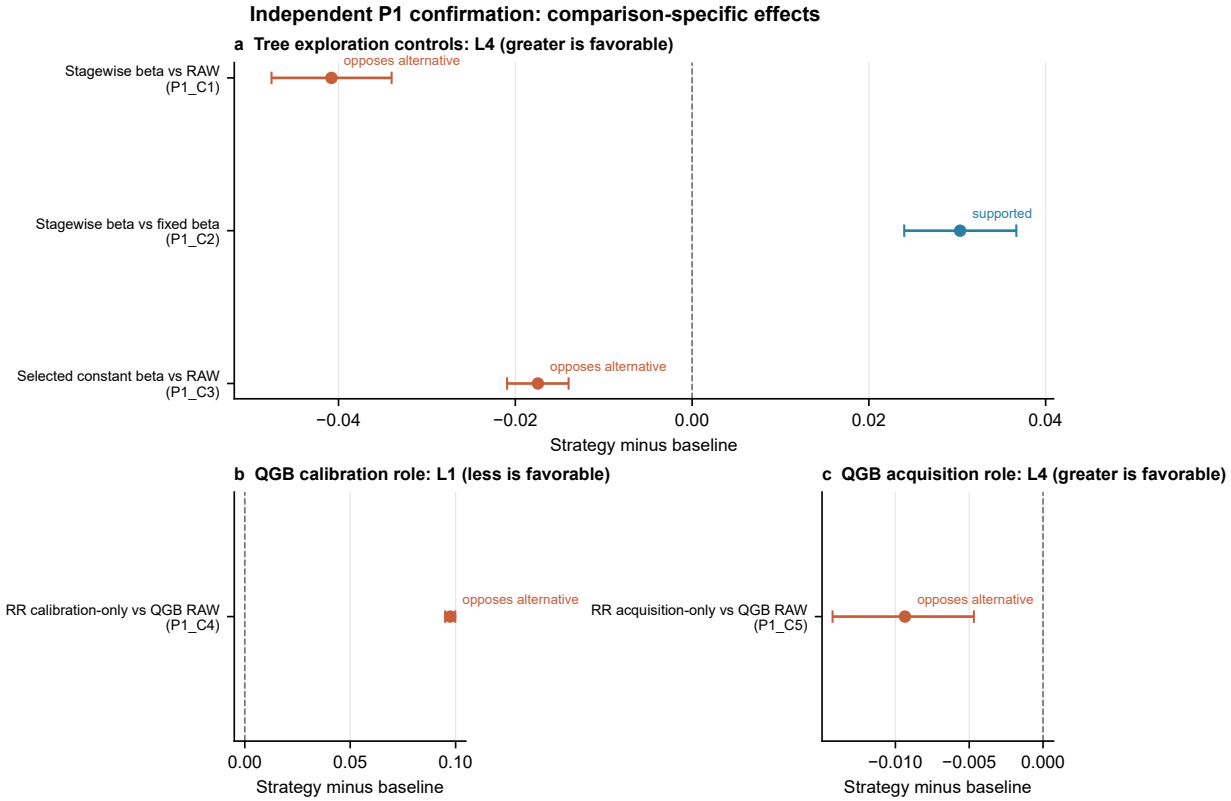


Figure S5: Registered P1 effects separated by endpoint and role; confirmatory display. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

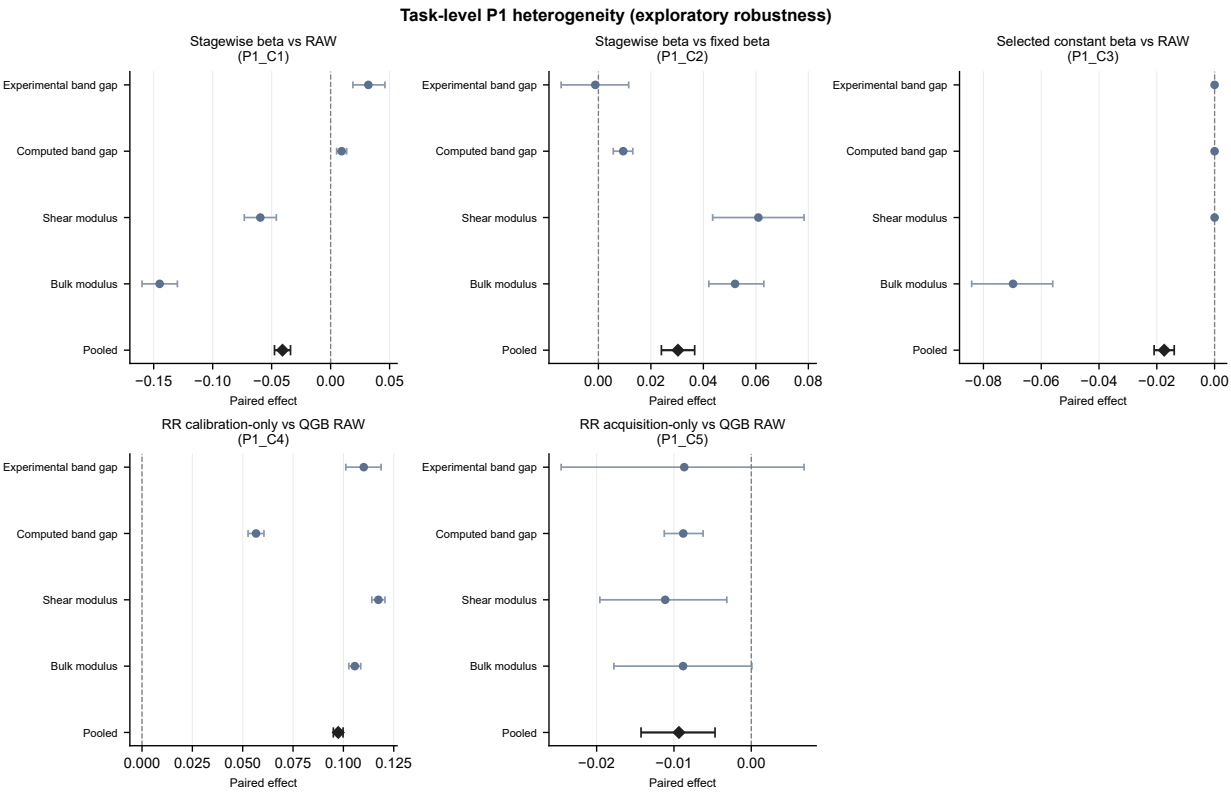


Figure S6: Four scientific-task effects plus pooled estimates; exploratory robustness. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

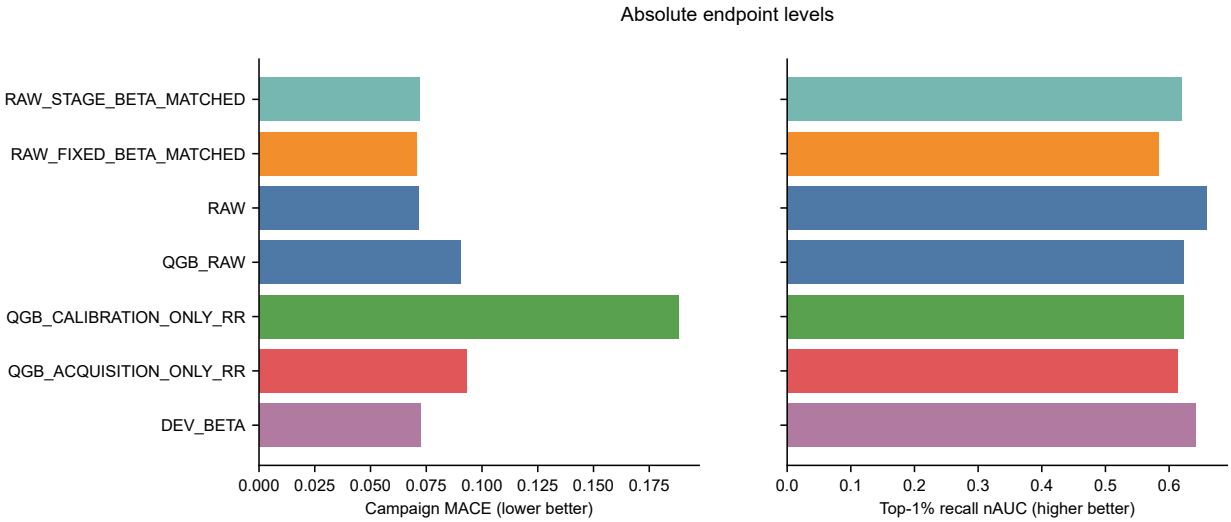


Figure S7: Absolute L1 and L4 levels; descriptive endpoints. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

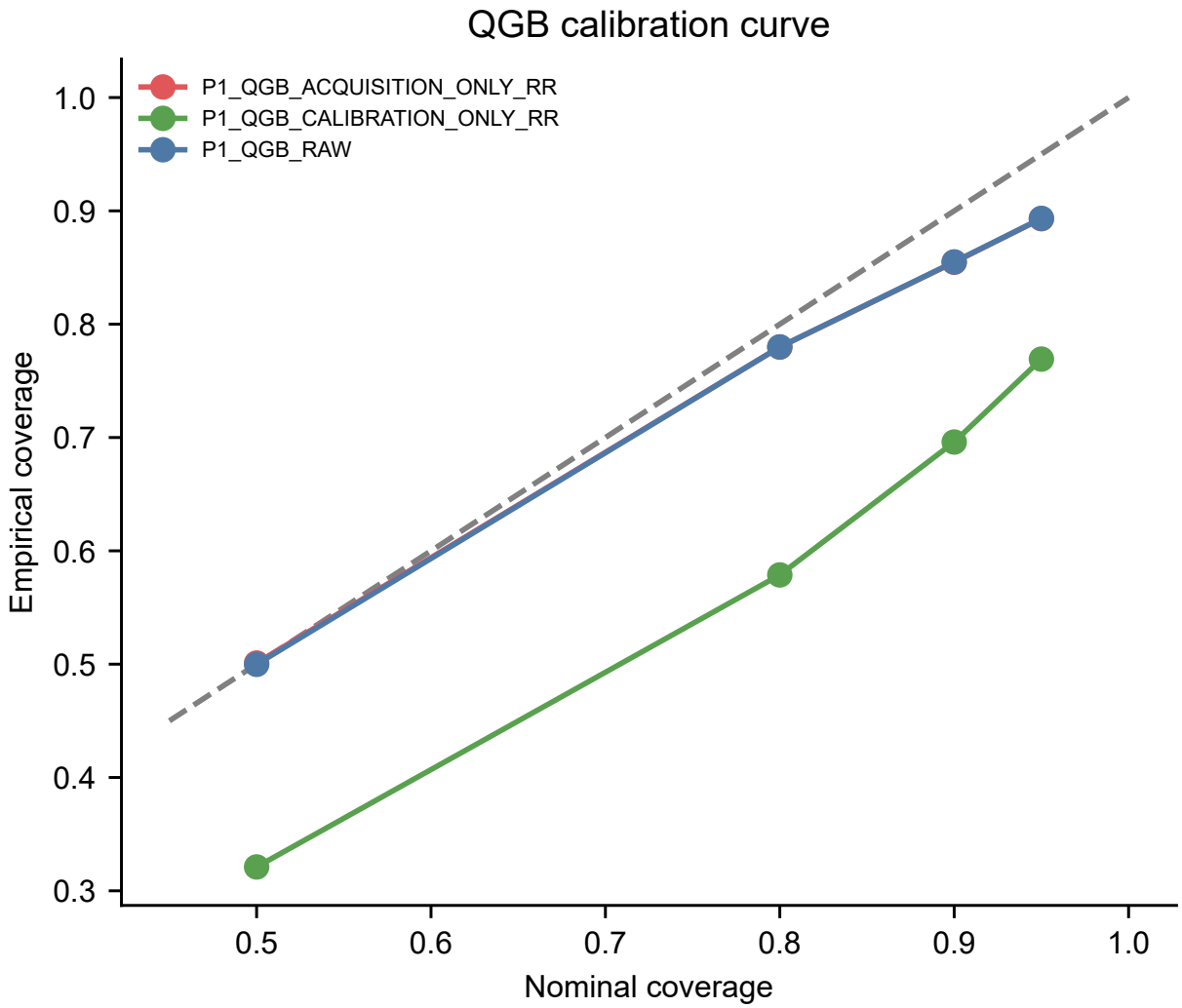


Figure S8: Nominal versus empirical coverage; exploratory diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

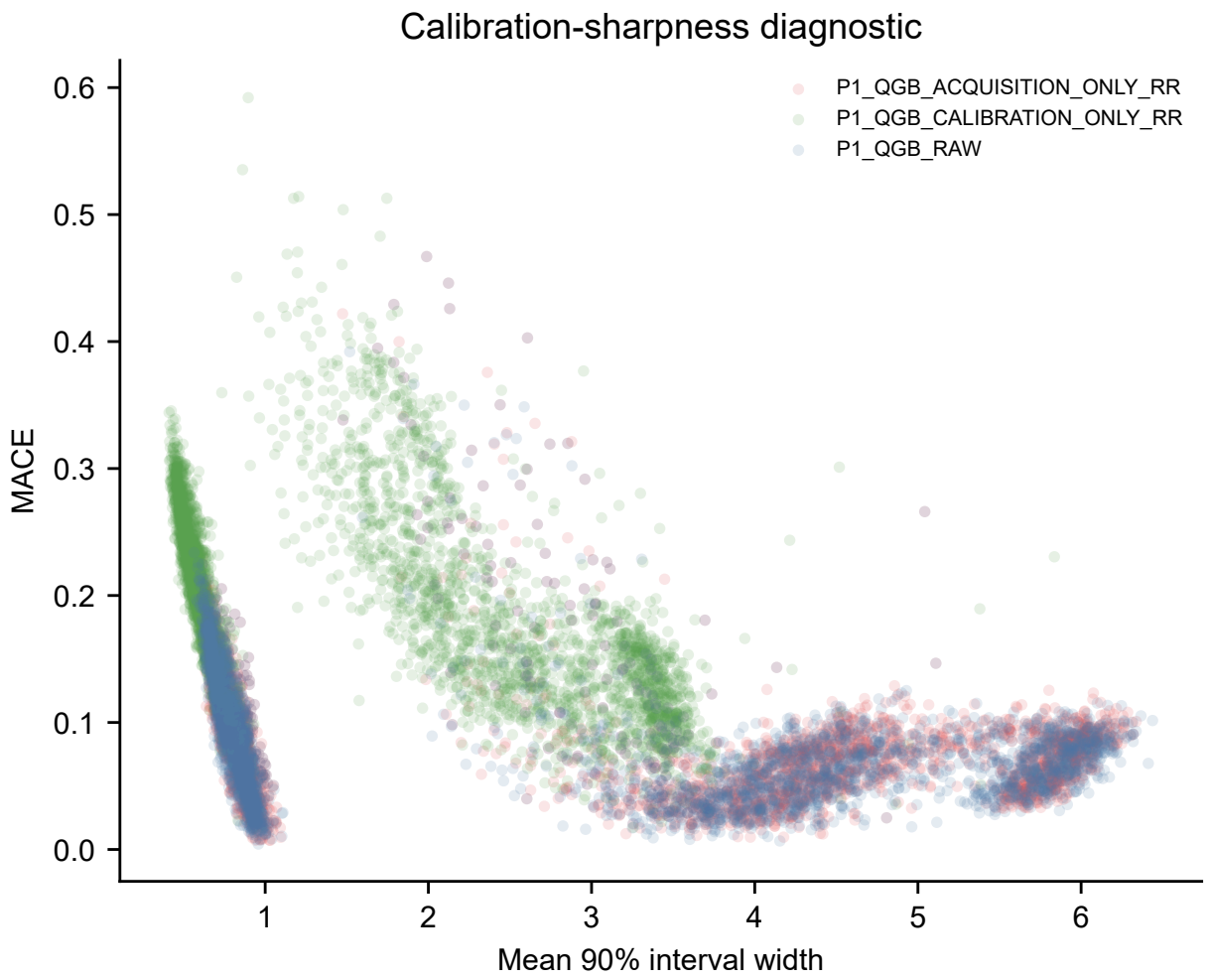


Figure S9: Interval width versus calibration error; exploratory diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

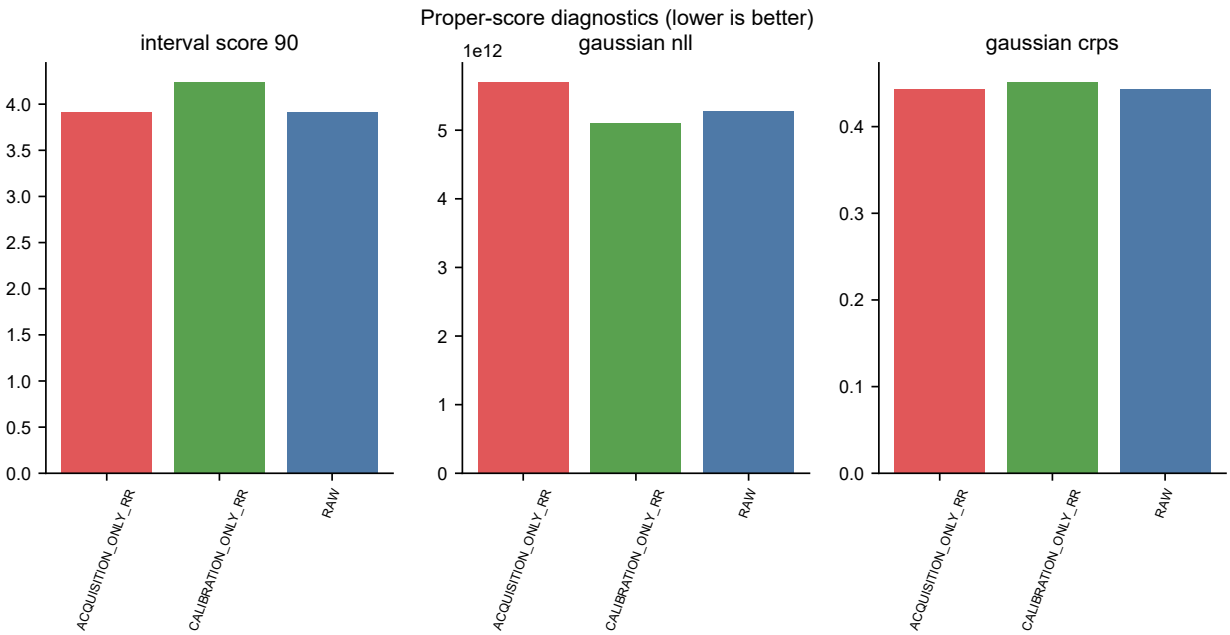


Figure S10: Interval score and Gaussian-proxy proper scores; exploratory diagnostics. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

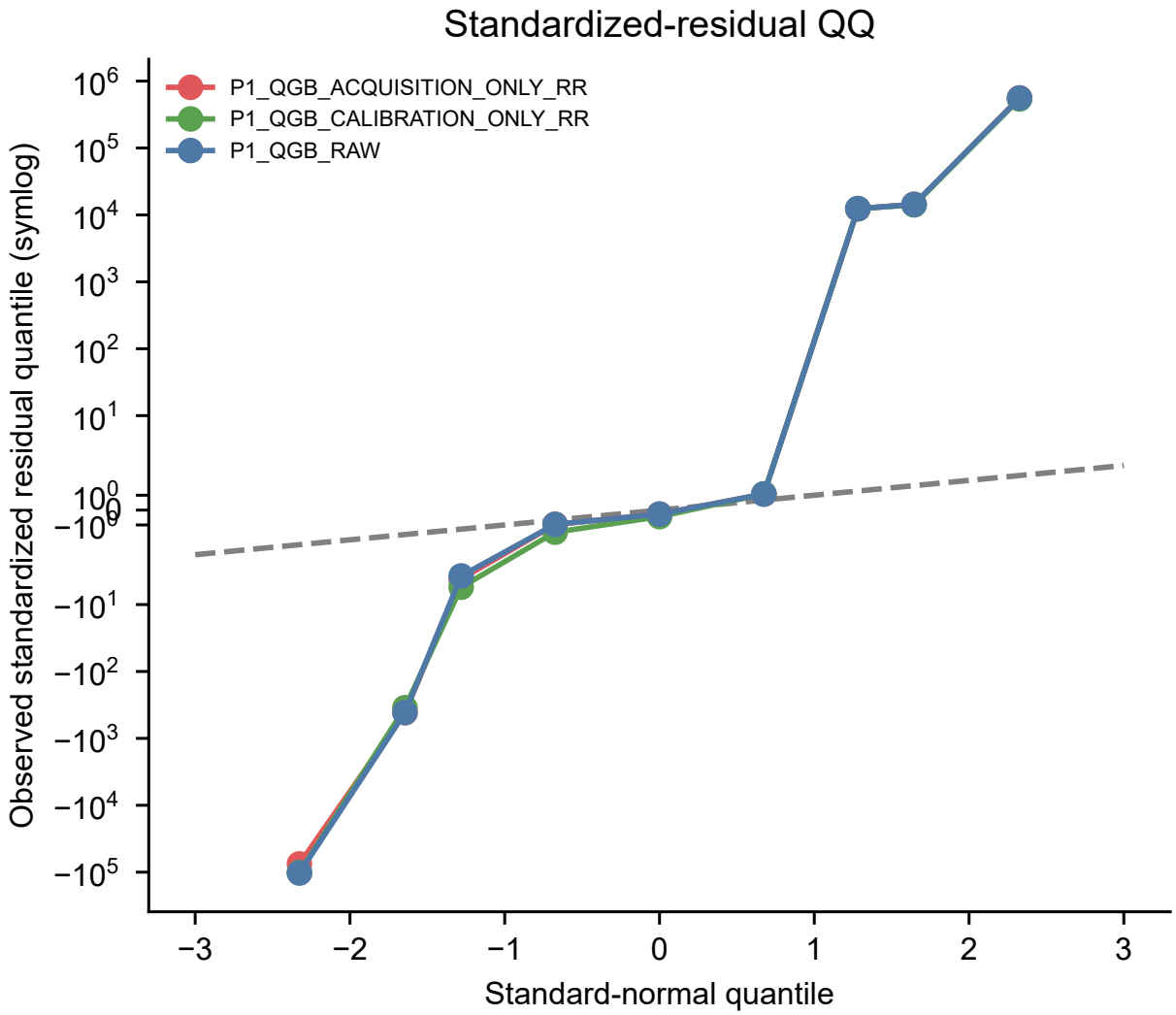


Figure S11: Standardized-residual QQ display; descriptive diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

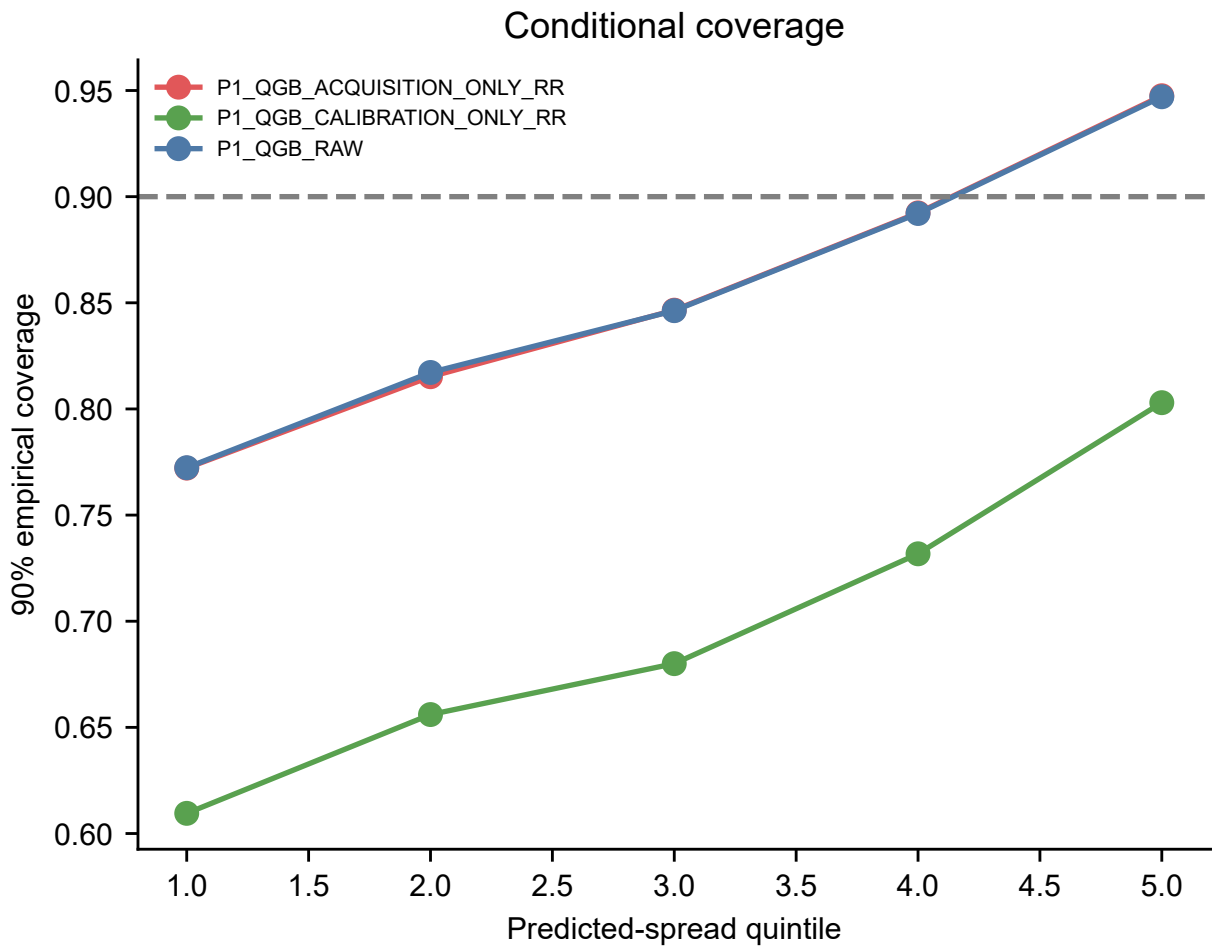


Figure S12: Conditional 90% coverage by predicted-spread quintile; exploratory diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

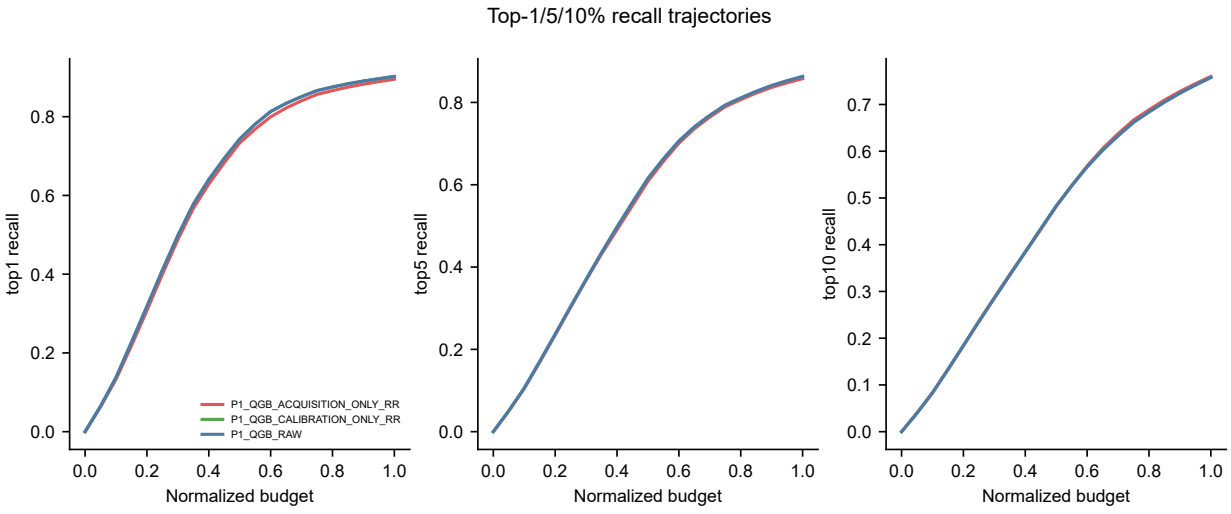


Figure S13: Top-1/5/10% recall trajectories; exploratory discovery diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

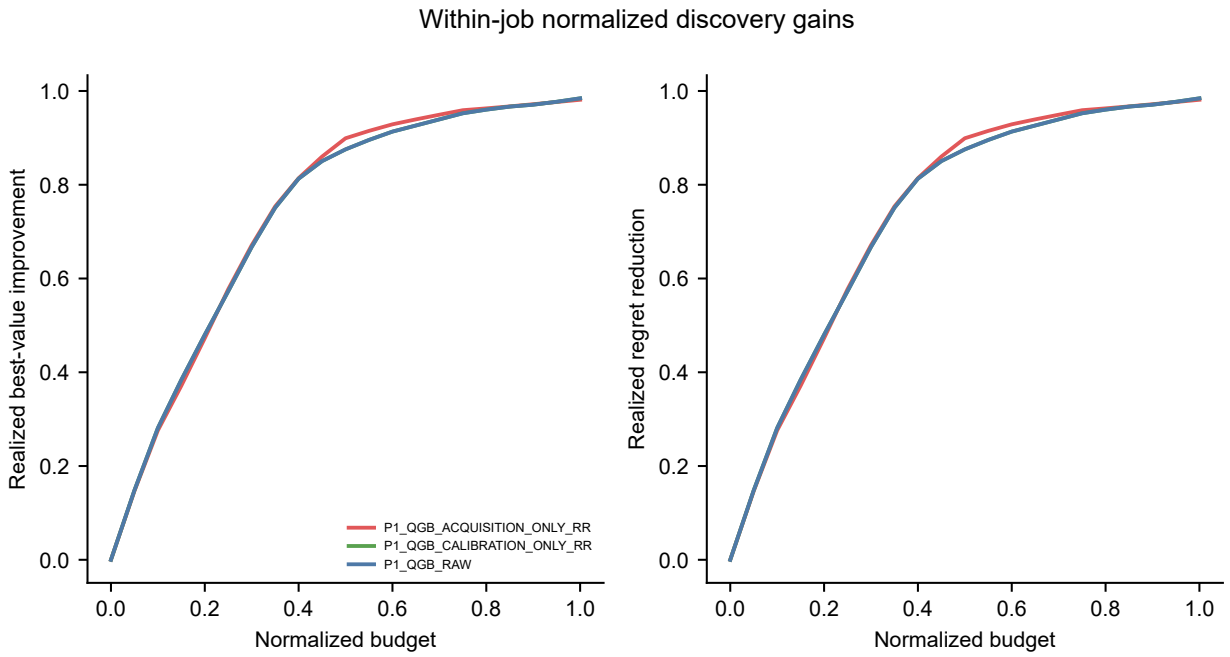


Figure S14: Within-job normalized best-value improvement and regret reduction; exploratory diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

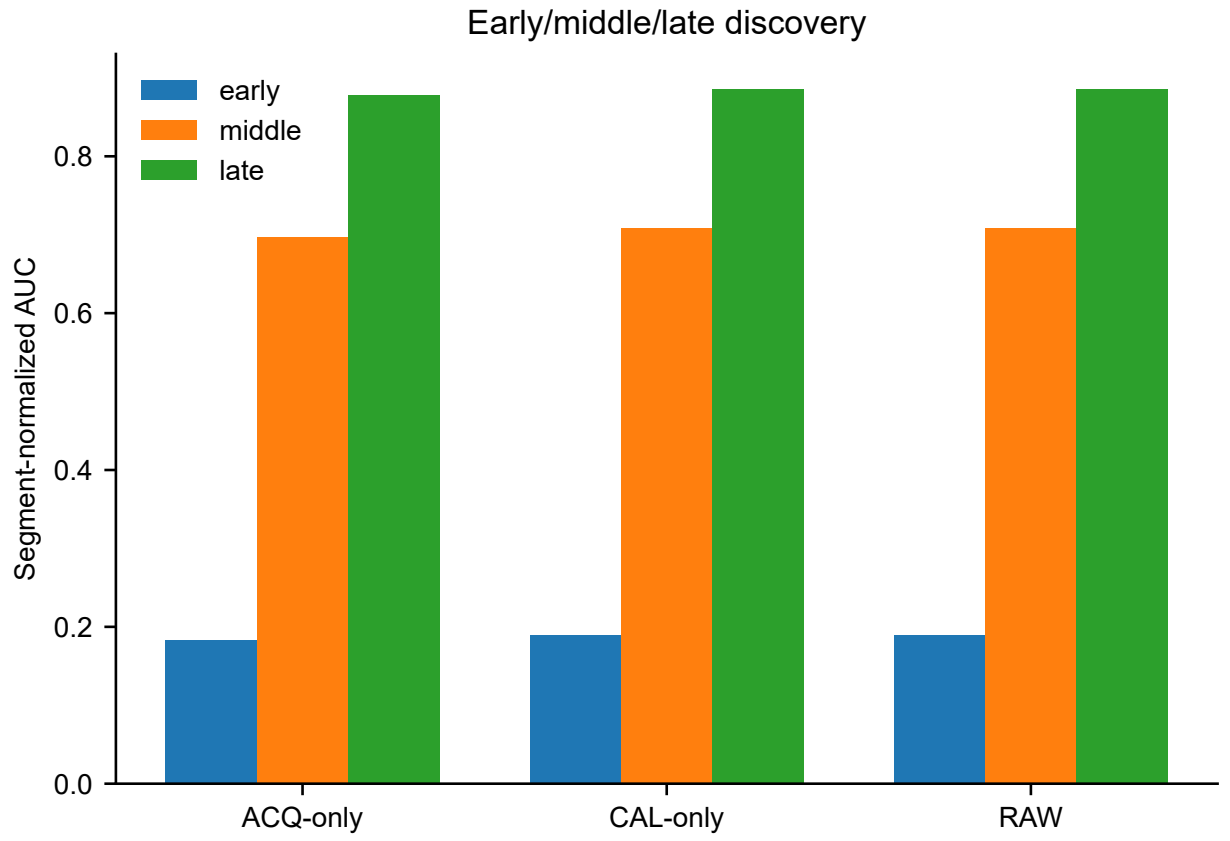


Figure S15: Early, middle and late normalized discovery AUC; exploratory diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

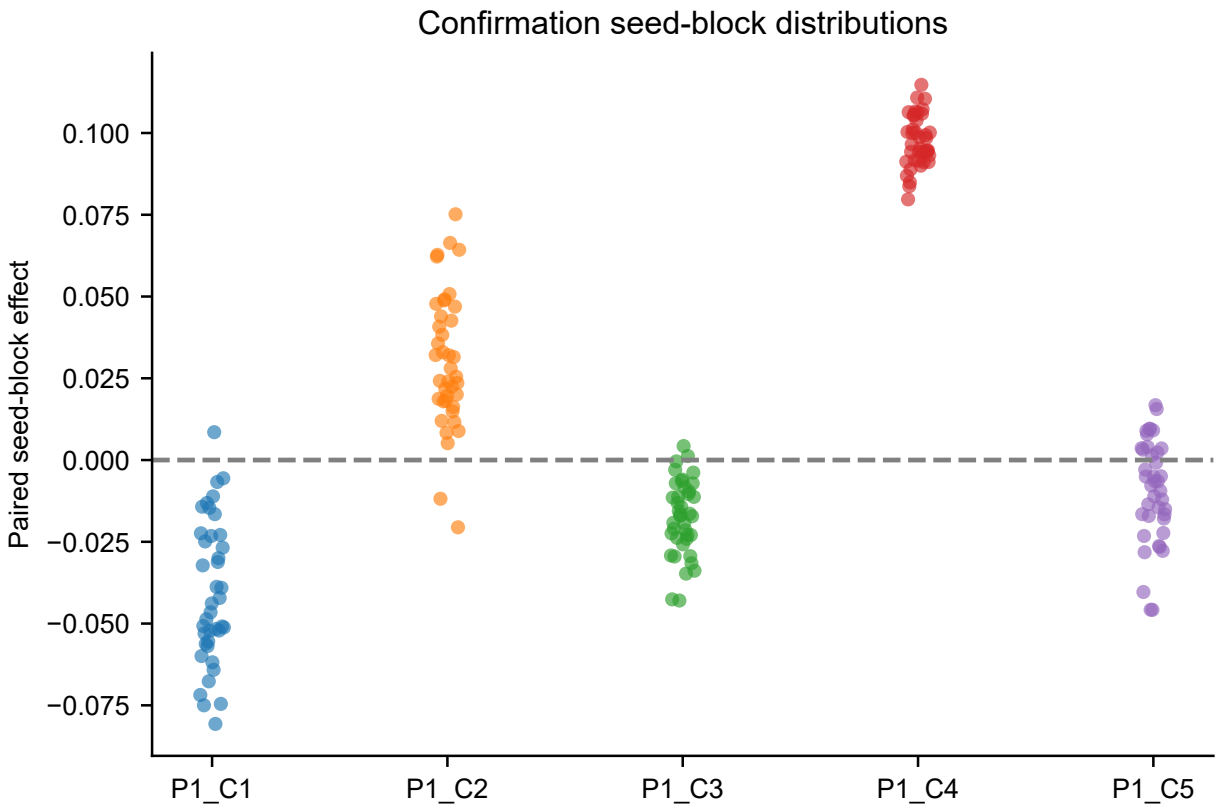


Figure S16: Paired confirmation seed-block effects; inferential units shown descriptively. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

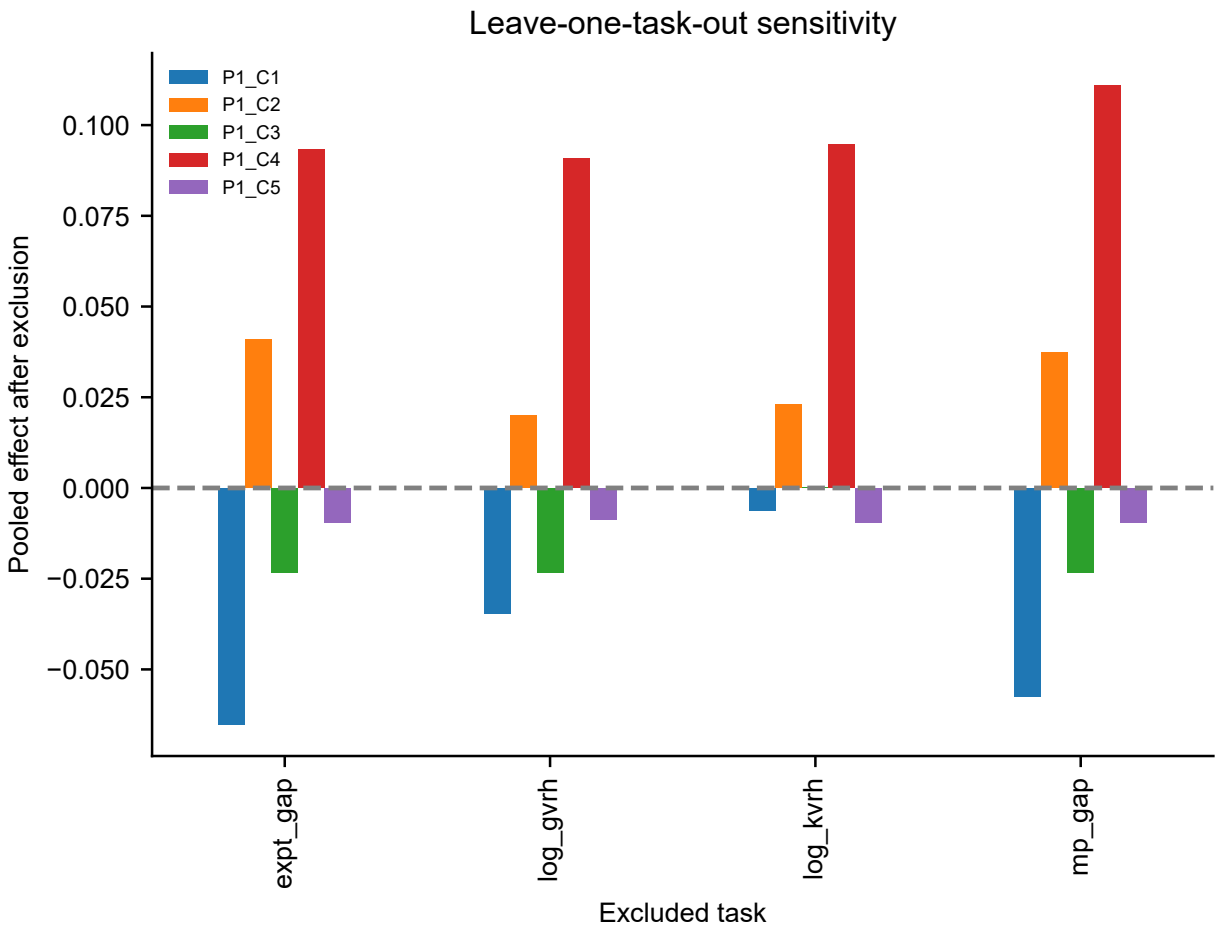


Figure S17: Leave-one-task-out pooled sensitivity; exploratory robustness. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

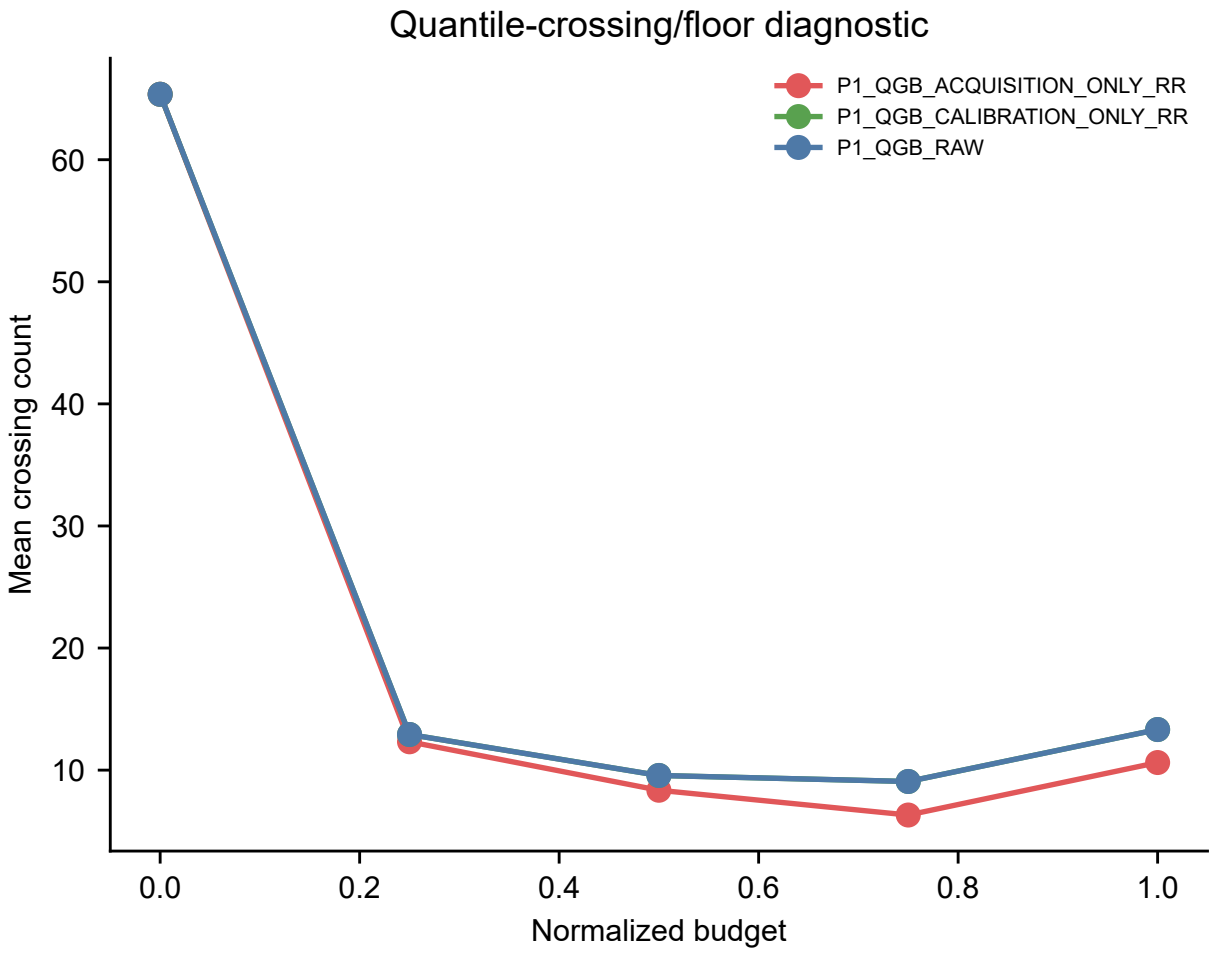


Figure S18: Quantile-crossing counts; descriptive implementation diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

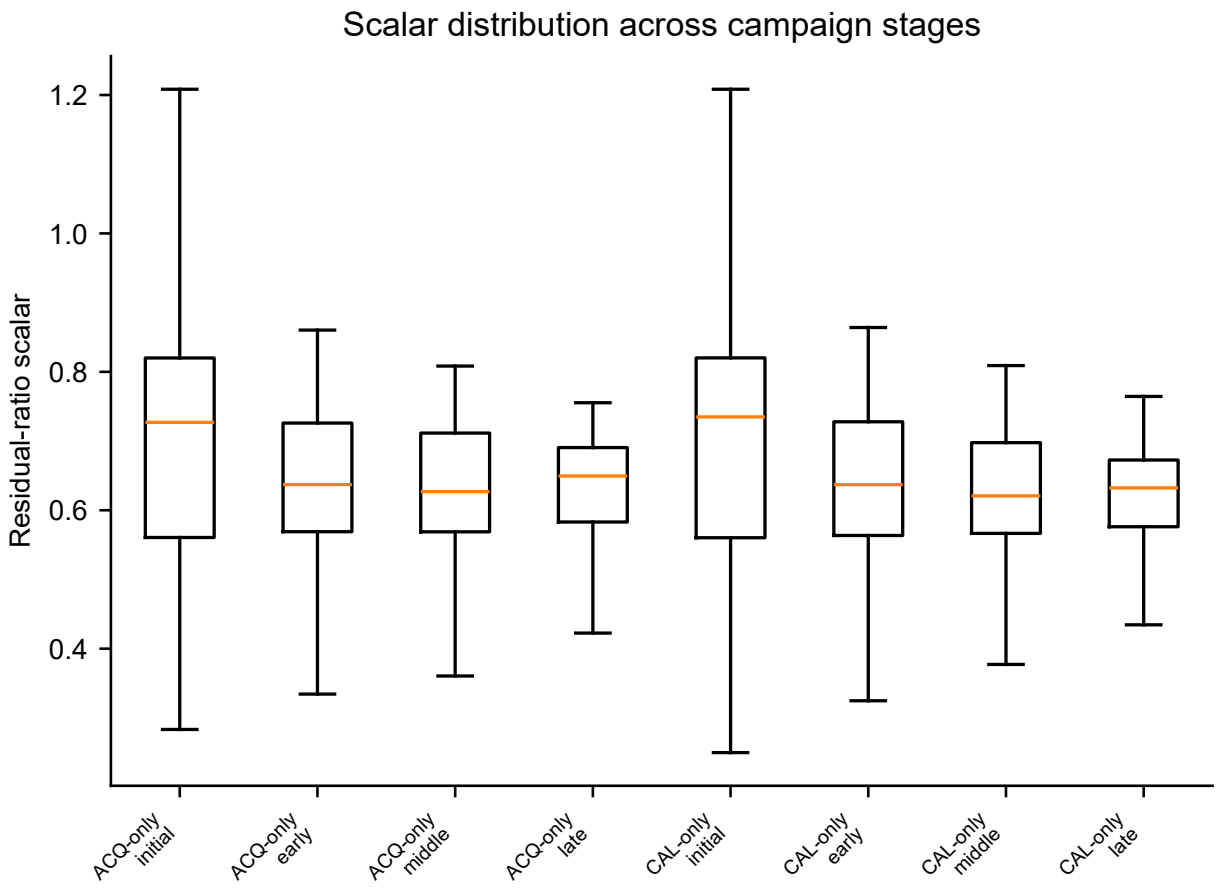


Figure S19: Residual-ratio scalar by campaign stage; descriptive diagnostic. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

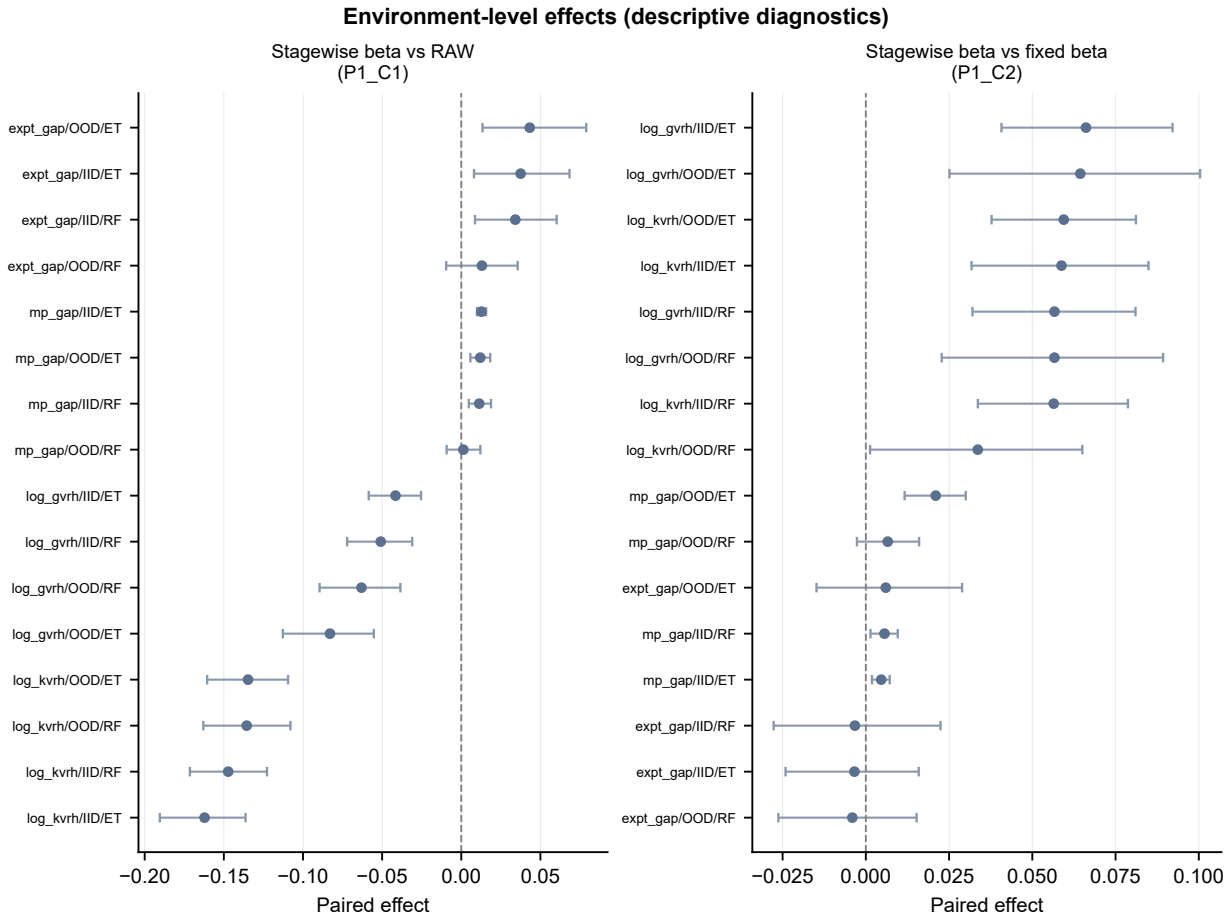


Figure S20: Environment-level P1_C1 and P1_C2 effects; descriptive diagnostics. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

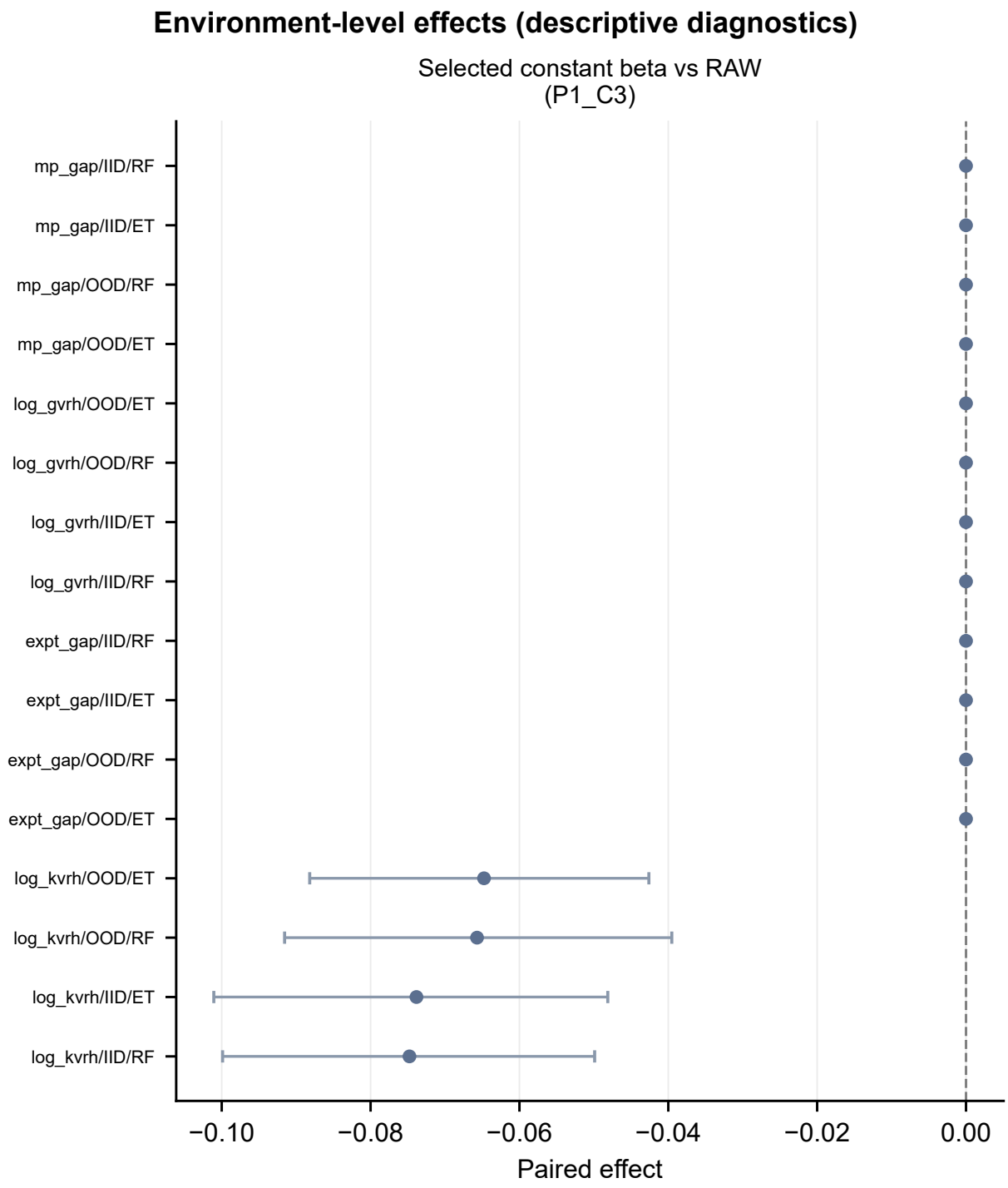


Figure S21: Environment-level P1_C3 effects; descriptive diagnostics. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.

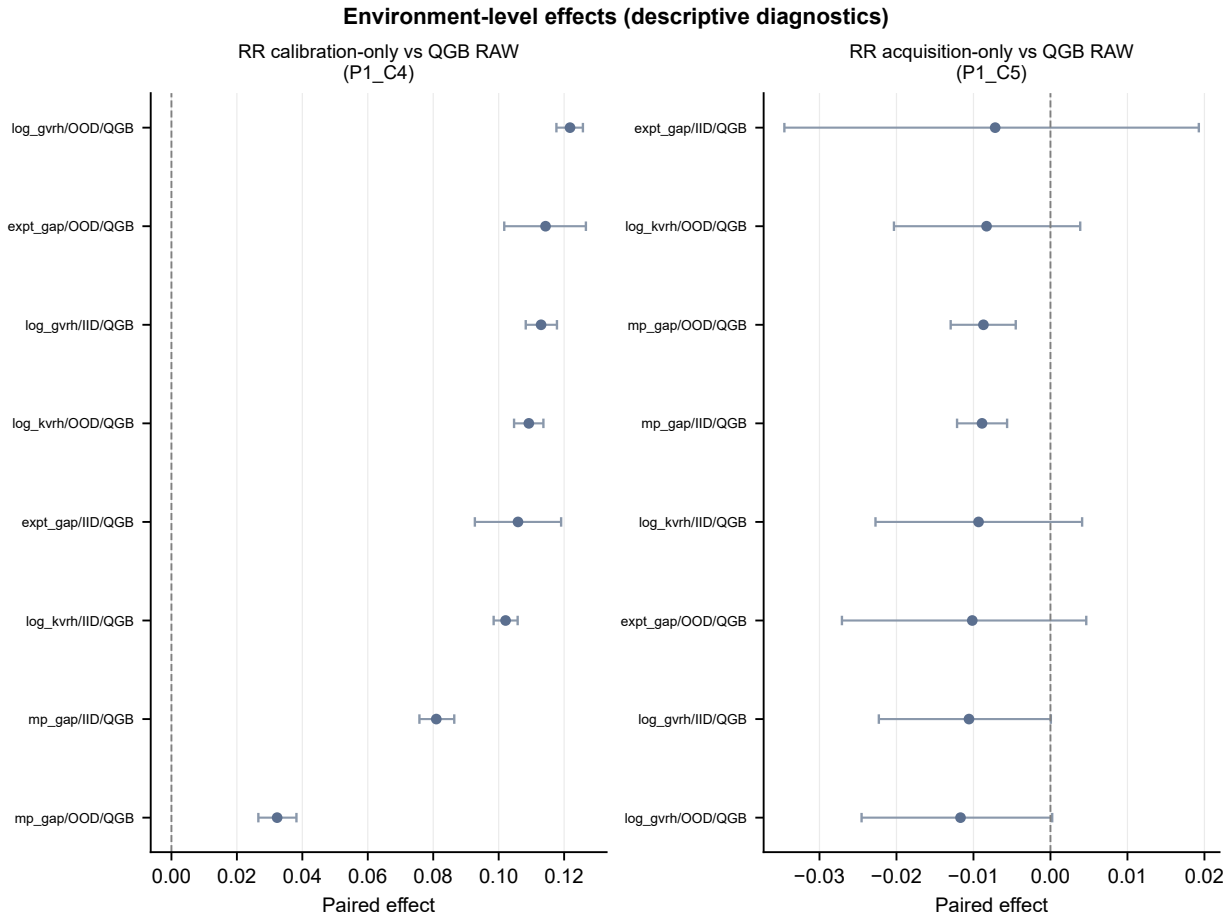


Figure S22: Environment-level P1_C4 and P1_C5 effects; descriptive diagnostics. Confirmation used seeds 8200–8239. Pooled multiplicity status is stated only for the registered-effect display.